
epiinfo™
Guide Series
Epi Info™ Guide to Epi Map

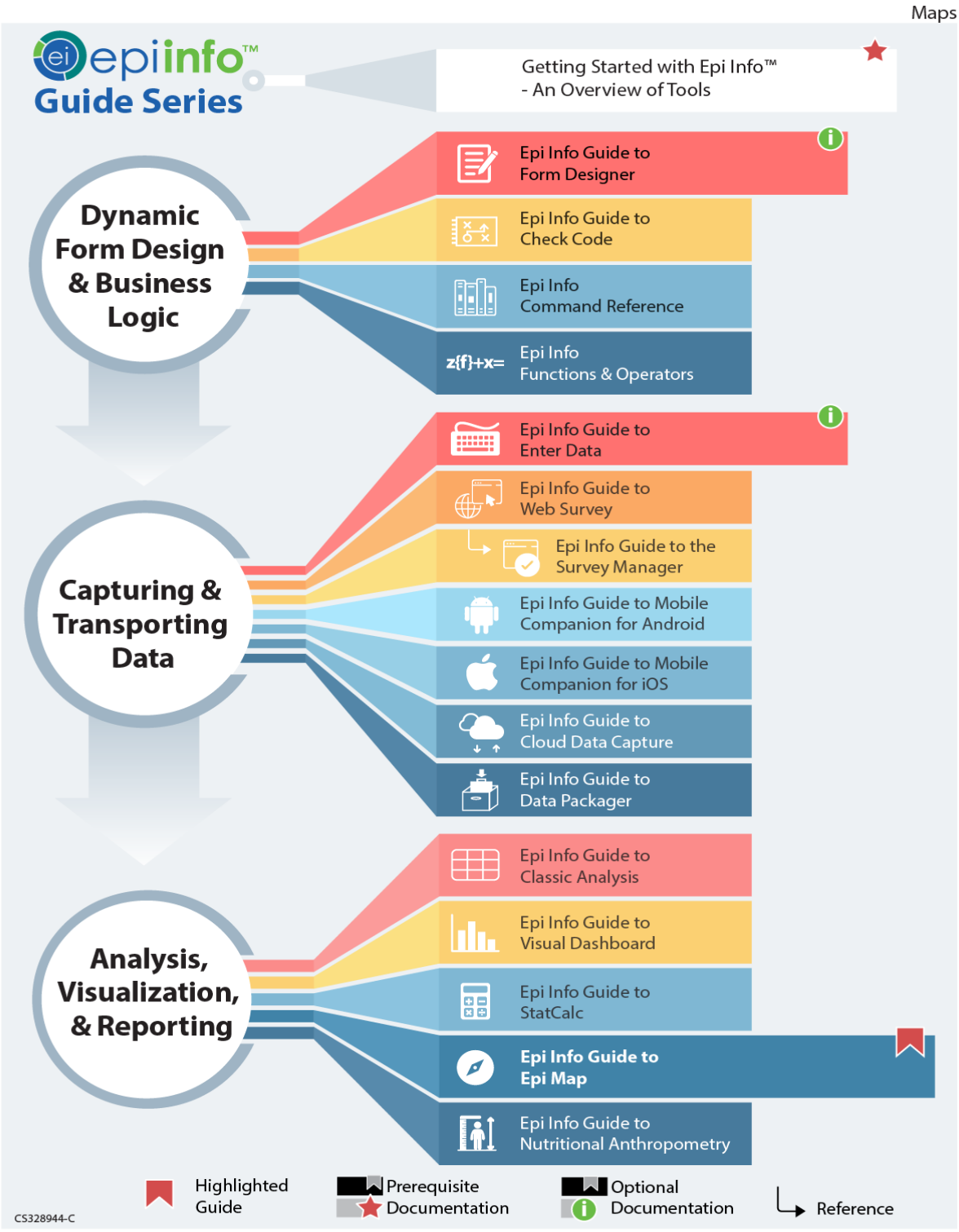
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EPI INFO™ 7

EPI MAP: Visual Representation of Data

Centers for Disease Control and Prevention

The image below shows the interrelationships among the guides in the series. This guide is *Epi Map*.



Preface

Epi Info™ 7 is a public domain suite of interoperable software tools designed for the global community of public health practitioners and researchers. Perform data analysis with epidemiologic statistics, maps, and graphs. Build data entry forms, construct a database, and customize statistics applications. Physicians, epidemiologists, and public health officials without a tech background can easily work with critical data using Epi Info™ tools. Epi Info™ 7 is free of charge. You can Download Epi Info™ 7 from the Centers for Disease Control and Prevention (CDC) website. There are many resources, components, and utilities in the Epi Info™ Suite of Tools. Each major tool has its own guide to help you make the best use of it. The graphic on the opposite page shows the full Epi Info™ Guide Series and the interrelationships between the guides. Each guide in the series has a similar graphic. The current guide is highlighted with a red bookmark icon (). This guide, ***Epi Info™ - Epi Maps: Visual Representation of Data by Location,*** contains a high-level description of each tool. Use the stand-alone guides for each Epi Info™ tool independently, or in conjunction with related guides for greater functionality. In some cases, related guides are important prerequisites to using the tool discussed in the current guide. The prerequisite documentation has a red star icon (). In other cases, you may find optional guides informative, and the optional documentation is shown with a green information icon (). For detailed instructions on using any of the Epi Info™ tools, refer to its specific guide in the Epi Info™ Guide Series.

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Epi Map: Visual Representation of Data by Location

Introduction

Epi Map is a tool in Epi Info™ 7 that helps you generate maps from your data. You can show your information in different ways on a map, implement filters to display specific criteria, and watch how your data changes over time. This makes it easier to understand public health information by seeing it on a map.

How Epi Maps Works

Your map displays information through layers that work together. The first type, called data layers, shows your information through case clusters, colored areas, or patterns of dots. The second type, called reference layers, adds helpful map features like boundaries, markers, and geographic details. You'll find tools at the bottom of your map window to change how your map looks and what information it displays.

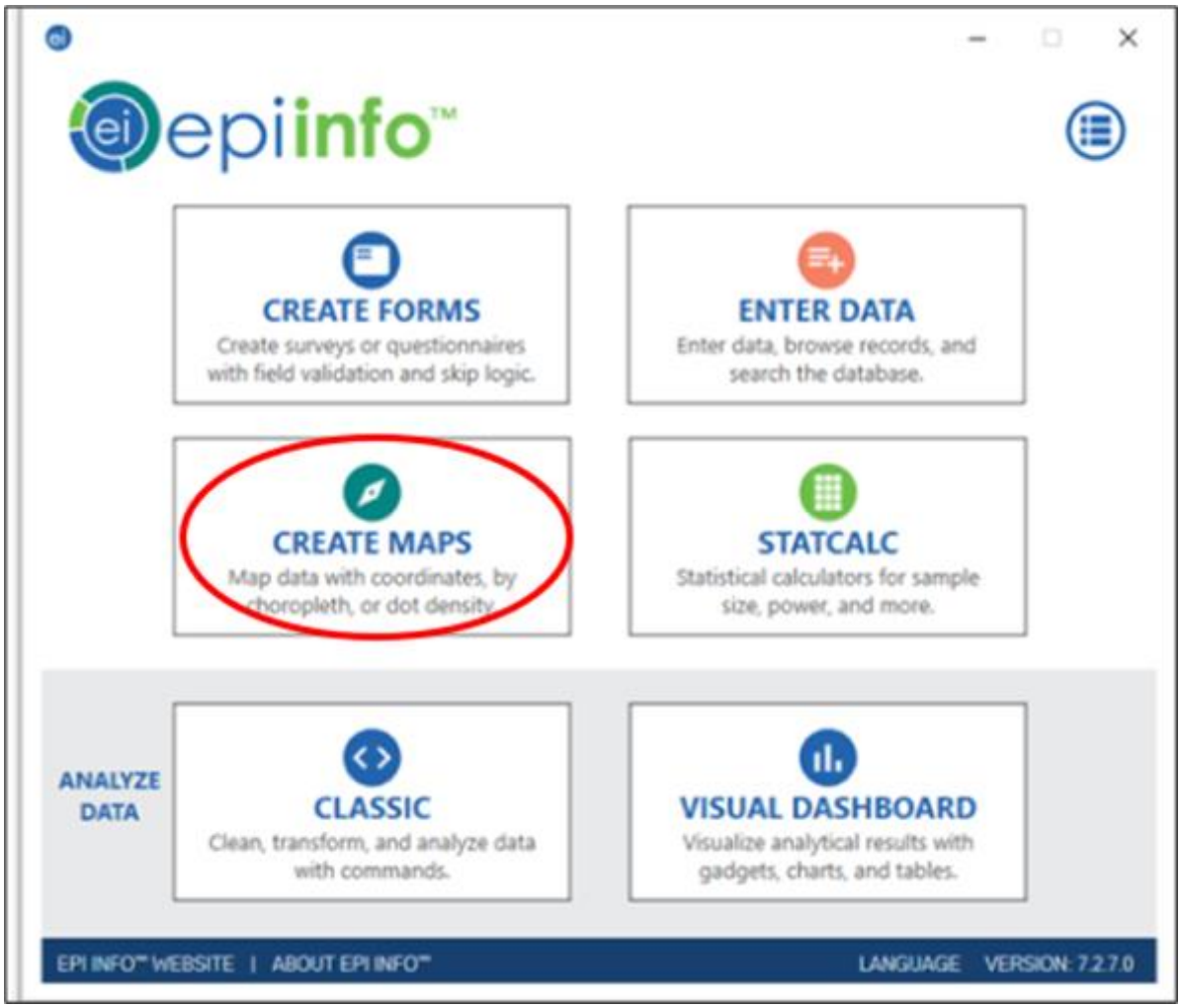
This organization of layers lets you build exactly the map you need. You can combine different types of information, add or remove details, and adjust how everything appears. The tools are designed to help you create clear, meaningful maps for your public health work.

Basic Tools

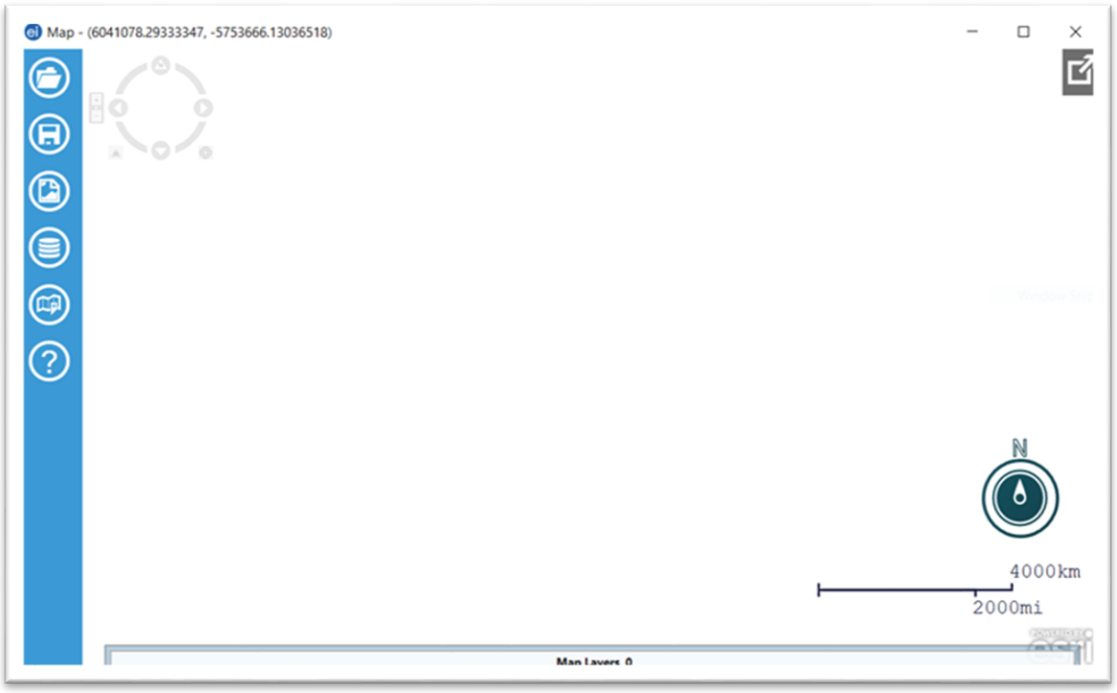
Opening Epi Map

You can open Epi Maps in two ways. The simplest way is to click Create Maps in the Epi Info™ 7 main menu. You can also open it by clicking Map in the toolbar at the top of the Enter Data screen. This guide explains how to use Epi Map when you open it from the main menu.

When you open Epi Map through the Enter Data toolbar, you get extra features that connect your project data with the mapping tool. To learn more about these additional features, you can read the Enter Data section of this guide.



Main Menu with Create Maps highlighted



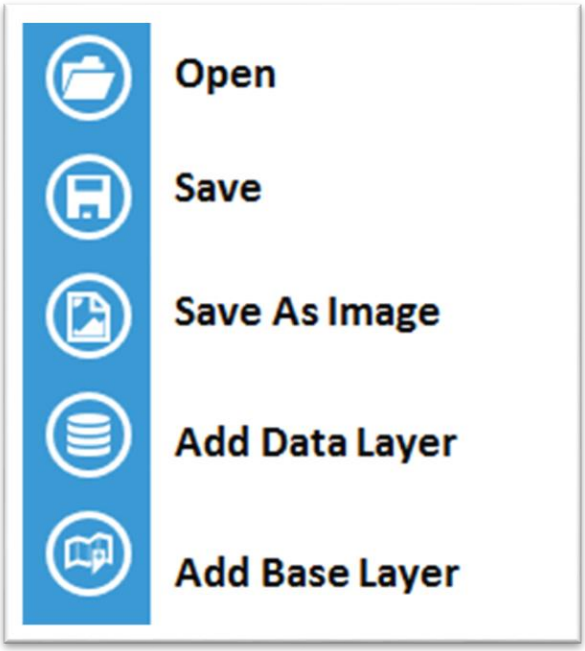
Error! No text of specified style in document.**Epi Map main workspace window**

View Settings

When you first open Epi Maps, you'll see a blank view. The blank view gives you an empty background, which works well when you want to add your own boundary files or base layers. Most of the work in Epi Map requires map layers to be constructed with shapefiles and related data variables or latitude and longitudes coordinates captured in a database. You will use the Add Data Layer functionality to add, remove, or re-order layers. Once a map is displayed, you will be able to filter certain criteria, add additional data, add points, , and labels to a map.

Maps can be saved as MAP files (interactive maps) that are editable and can be opened in Epi Map. Maps can also be saved as a portable network graphic PNG file and used in presentations and in other programs.

Other features are also accessible through the Epi Map menu bar. The following options are available on the Epi Map menu.



Error! No text of specified style in document.

Epi Map toolbar with icons defined

Open a map file

Interactive maps created and saved in Epi Map can be opened using this menu option.

Save file as a Map

This option allows users to save current maps as an interactive map. Users will be able to open this file as many times as needed without having to select data sources and regenerate data layers. If the data source is updated, changes will be reflected in the map.

Save file as an Image

This option allows users to save the map displayed in the current Epi Map session as a portable network graphic PNG file.

Add data layer

Allows you to incorporate different types of data/information into one single map.

Add base layer

Allows you to add a reference layer without any associated data needed.

Create time lapse

Based on a date field from your dataset, this feature allows you to plot points into your map using a motion effect to represent the order of occurrence based on the value of a date field. This feature can only be used for maps where only one case cluster layer with latitude and longitude has been loaded.

Map Types

Epi Map lets you add different layers of data to your map. Think of layers like transparent sheets stacked on top of each other, each showing different information. You can choose from four types of maps that can be generated using data layers.

Choropleth

Use to categorize features into equal ranges or counts (quantiles) that are color-coded to indicate changes in the data. Using this guide, you will create a choropleth map that illustrates income data for the region.

Case Cluster

Feature allows users to show exact locations of people with a specific attribute. Using this guide, you will create a case cluster map that shows the physical household locations of the children interviewed based on addresses captured for each household. You will also draw the locations of hospitals and schools based on latitude and longitude data.

Dot Density

A map type that uses dots or points to show a comparative density of features over a map based on values stored in polygon layer attribute fields. Dot density is an effective method to visualize concentrations of quantitative data. In a dot density map, the dots are all the same size and do not represent actual point locations of data.

Spot Map

A map type that uses a symbol (diamond, square, cross) to represent the location of a phenomena or point of interest (for example location of water pumps near Broad Street). Spot maps require latitude and longitude values corresponding to the location(s) to be drawn into the map. Using this guide, you will incorporate spot maps to draw the locations of hospitals and schools based on latitude and longitude data.

Creating a Choropleth Map

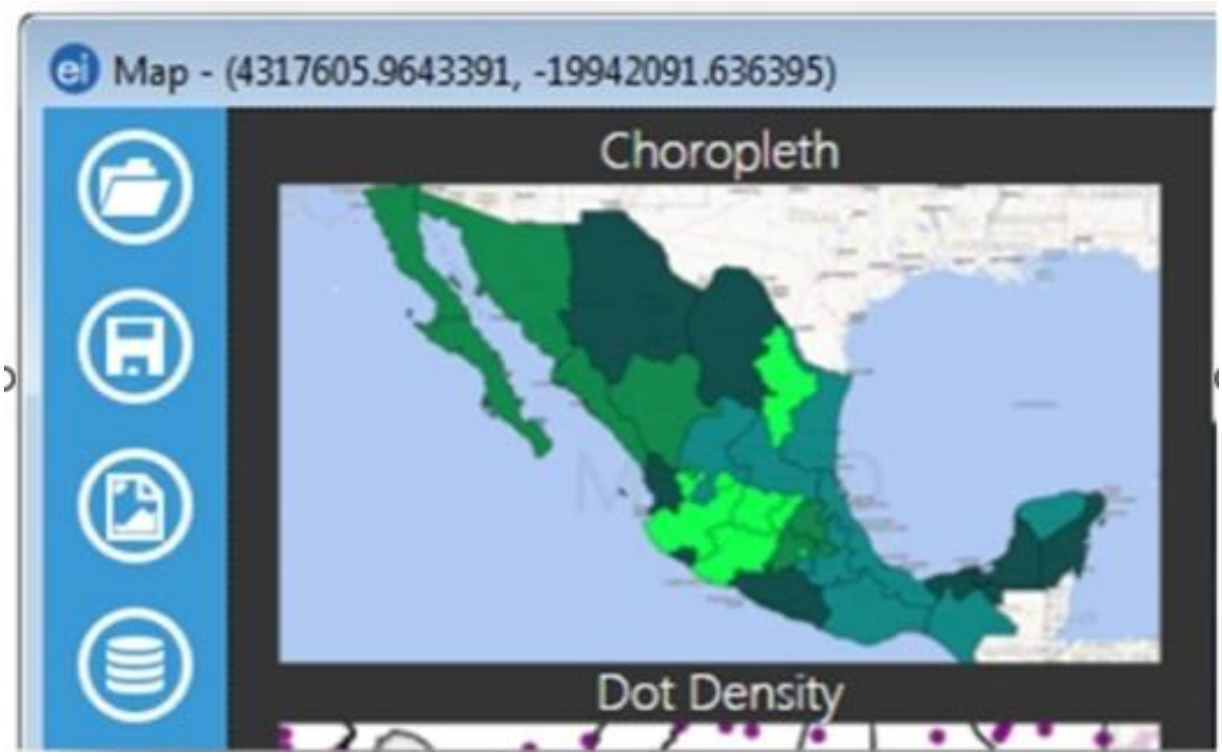
A choropleth map is a type of thematic map that uses color or shading to represent the value of a specific variable across different geographic regions. It visually conveys data such as population density or average income by coloring predefined areas based on data. To create a Choropleth map, complete the following steps:

1. Open Epi Map.
2. Click the **Add Data Layer** menu option.



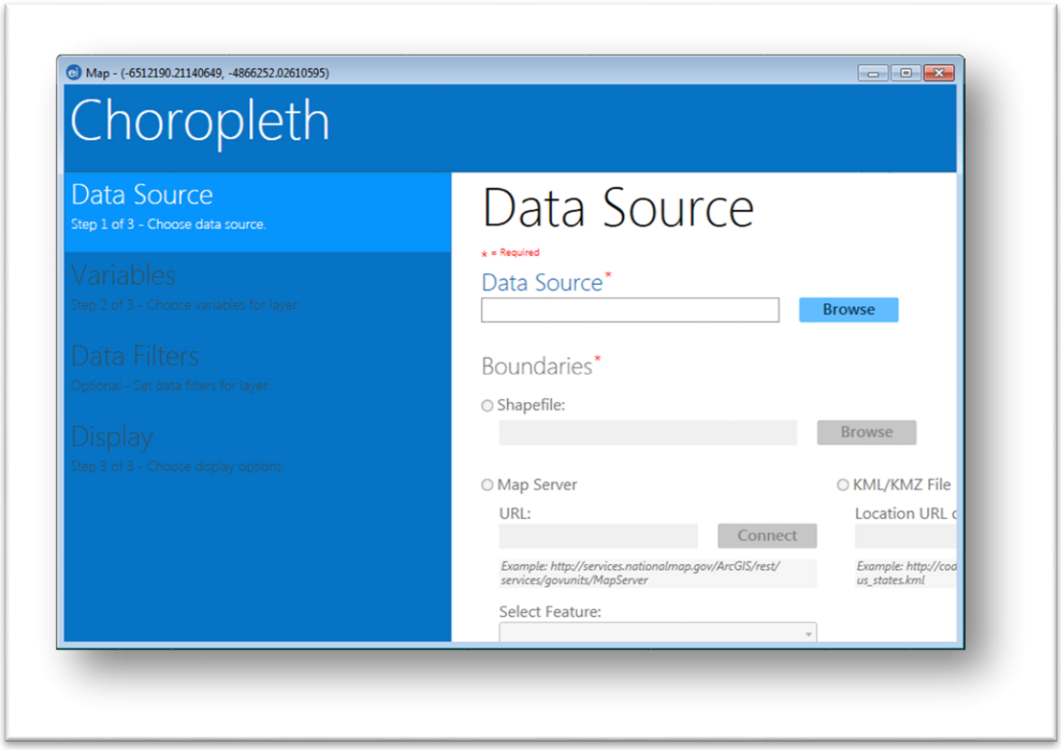
Epi Map workspace

3. Click on **Choropleth**.



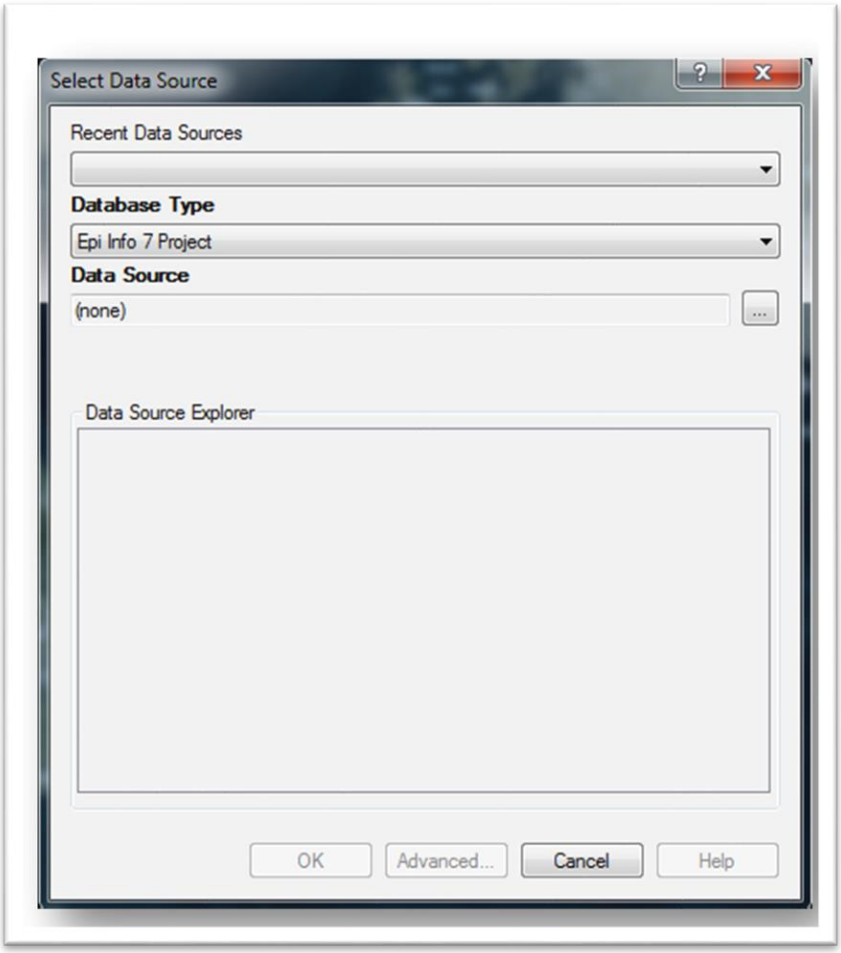
Epi Map Choropleth map option

4. The Choropleth Map Configuration window opens.



Epi Map Choropleth Data Source

5. Click on the **Browse** button. The Select Data Source window opens.

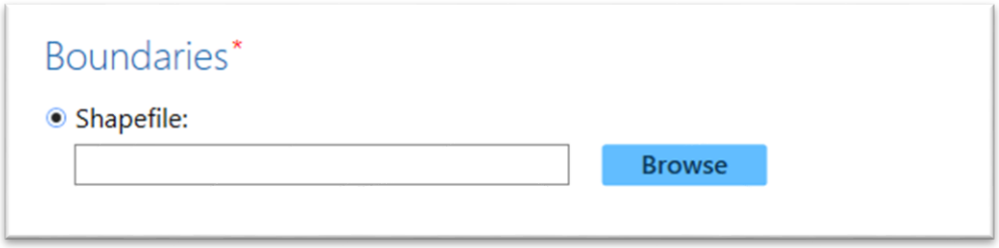


Select Data Source window

5. Select **MS Access 200-2003** as the Database Type.
6. Locate the file **albany_demogz.mdb** file in the Epi Info/Projects/Community Health Assessment folder.
7. Click **Open**. The Select Data Table From dialog box opens.

- The demographic project has two tables. One contains income data and the other race and ethnicity data. You want to create a choropleth map using the income data. The income table contains a series of zip codes and the median income and per capita income for those zip codes. The median income is the value of family income with 50% of families above the value and 50% of the families below the value. Per capita income is the total income for that zip code divided by the population.

8. Select **INCOME**.
9. Click **OK**.
10. Click on the **Browse** button next to the Shape File radio button in the Boundaries section.



Shapefile option

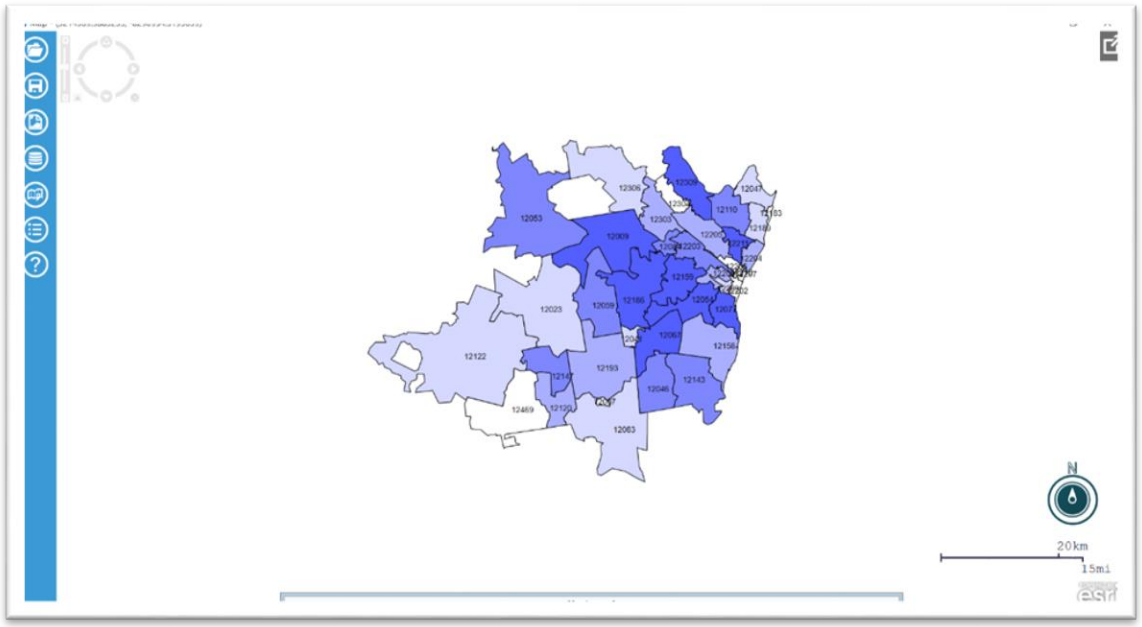
11. Locate the file **ALBZCTA_region.shp** in the Epi Info 7/Projects/Community Health Assessment folder.
12. Click **Open**.
13. In the Data Source section, select **ZIP** from the Data Key Field drop down list.
14. This is the field inside your data that matches the shape file.
15. In the Data Source section, select **MEDIANFAMILY** from the Value Field drop down list,
16. This is the data field to be mapped.

Note: *Information on zip code 12007 is not contained in the demographic project file.*

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17. From the Boundaries section, select **Zip** from the Feature Key Field drop down list.
18. This is the field inside the shape file that matches your data.
19. Keep the number of classes at **5**. The number of classes shows how many divisions or groups into which your data will be divided on the map.
20. Click **OK**. The choropleth map appears. Your map should look like the one below.

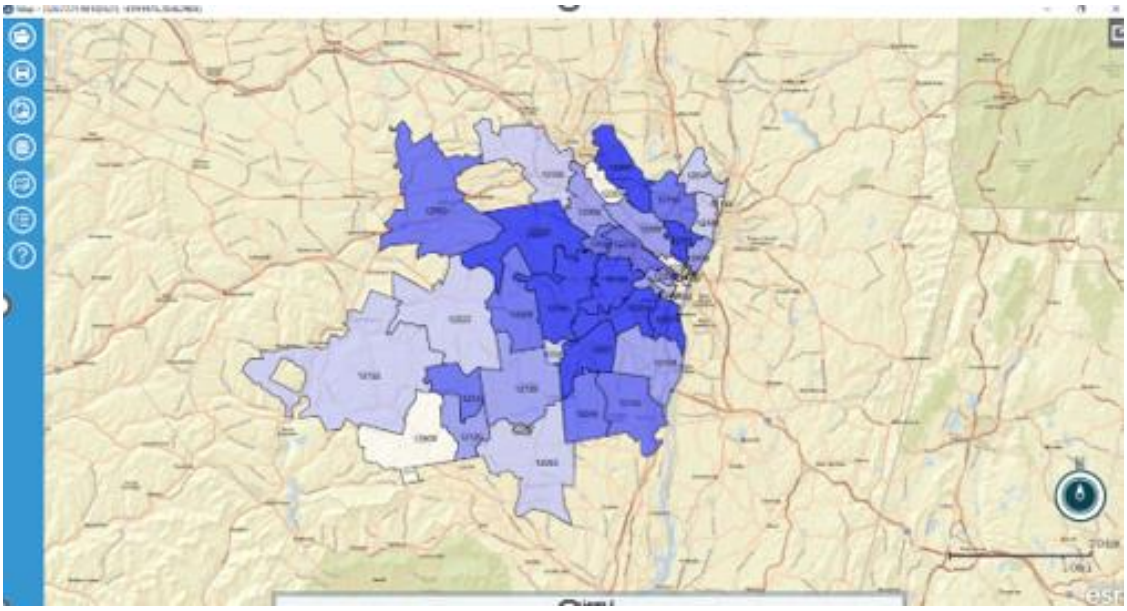
21.



Epi Map Choropleth map

To view a background image on your map you will need to use the Add Base Layer option. To add this layer, complete the following steps:

1. Select the **Add Base Layer** option.
2. Select **Map Server**
3. Paste and copy the provided with Epi Map (https://services.arcgisonline.com/ArcGIS/rest/services/World_Street_Map/MapServer)
4. Click **Connect**
5. Click **OK**. The choropleth map appears. Your map should look like the one below.

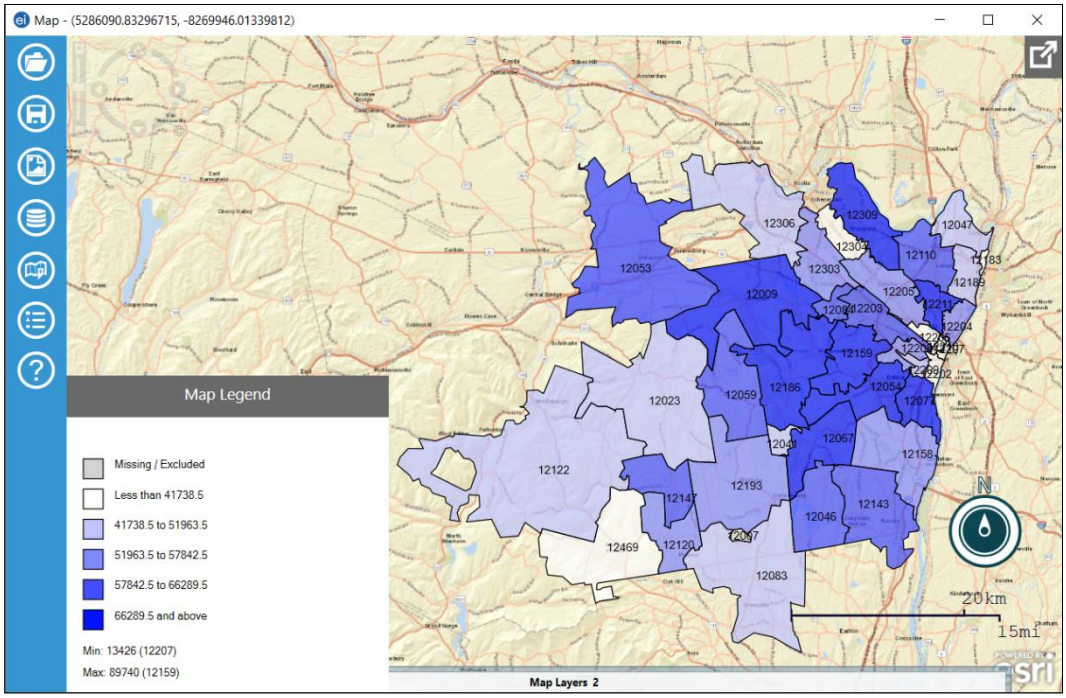


Choropleth map with base layer

Notice that there is no legend displayed when the map is first generated. To display the legend, click on Hide/Show Legend icon on the menu bar. The legend contains color codes for the median family income per zip code. The darkest color represents the highest income range, and the lightest color represents the lowest. Click the **Hide/Show Legend** button to display the legend.



Legend menu option



Choropleth map with legend

Changing Map properties

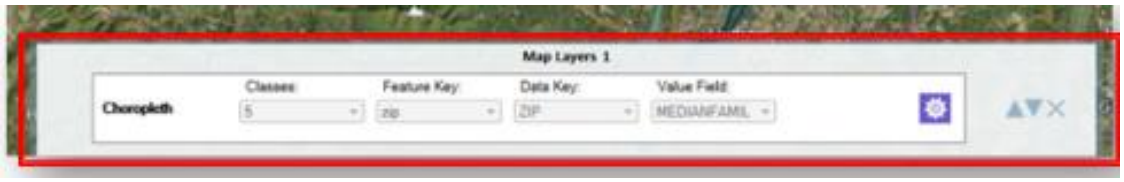
Notice that your legend contains a breakdown on information based on five divisions of data. To extend the ranges and further customize the map, you can complete the following.

1. To view the information on the current layer, place your cursor around the Map Layer slide-out gadget located in the bottom section of the screen.



Map Layer Slide-out

2. To change the number of classes, click on the gear icon.



Map Layer Gear

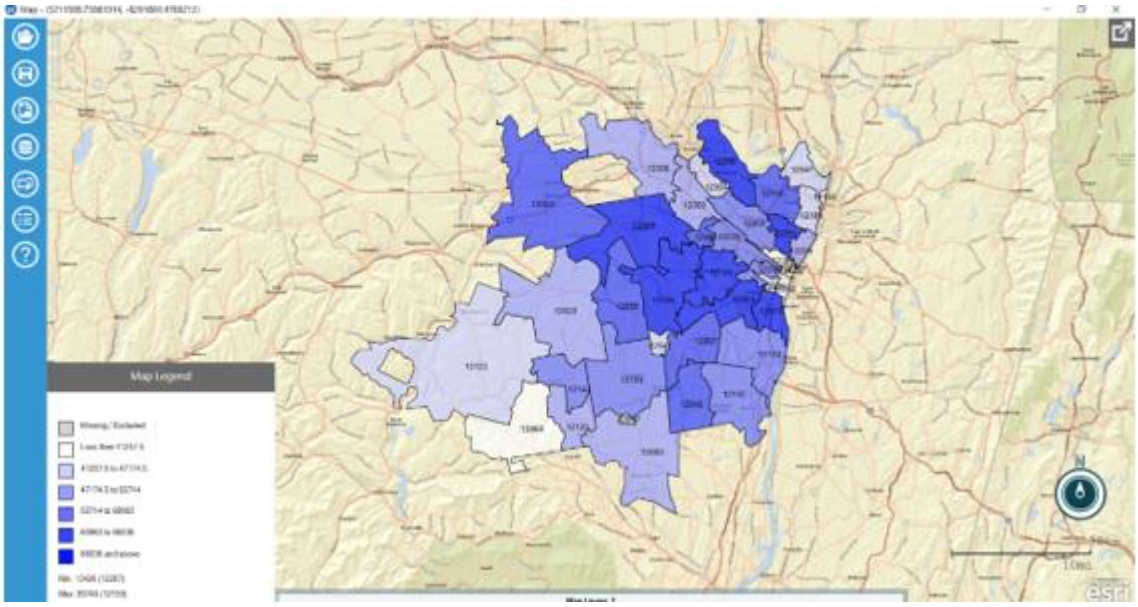
3. From the Classes drop-down list, select **6**. As mentioned before, the number of classes listed shows how many divisions or groups into which your data will be divided on the map.

    22. The color legend on the bottom shows the lightest to darkest colors for the map codes and allows changing the colors on the map. You

can use the color dialog box to change colors inside the map.

Experiment with the different colors.

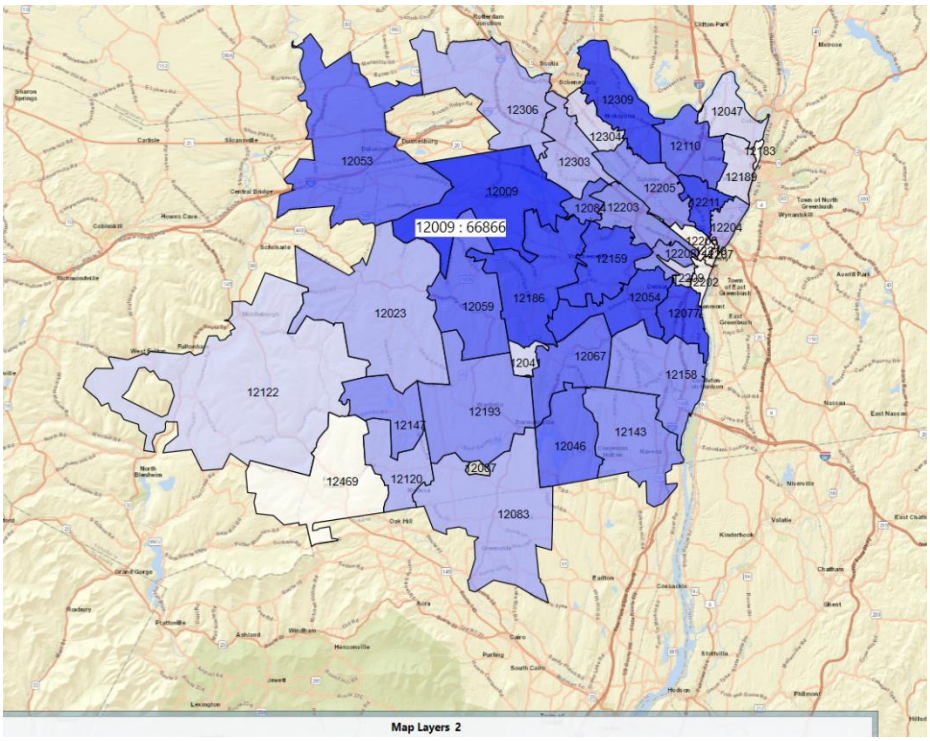
23. Your map should look like the following:



Map view with legend

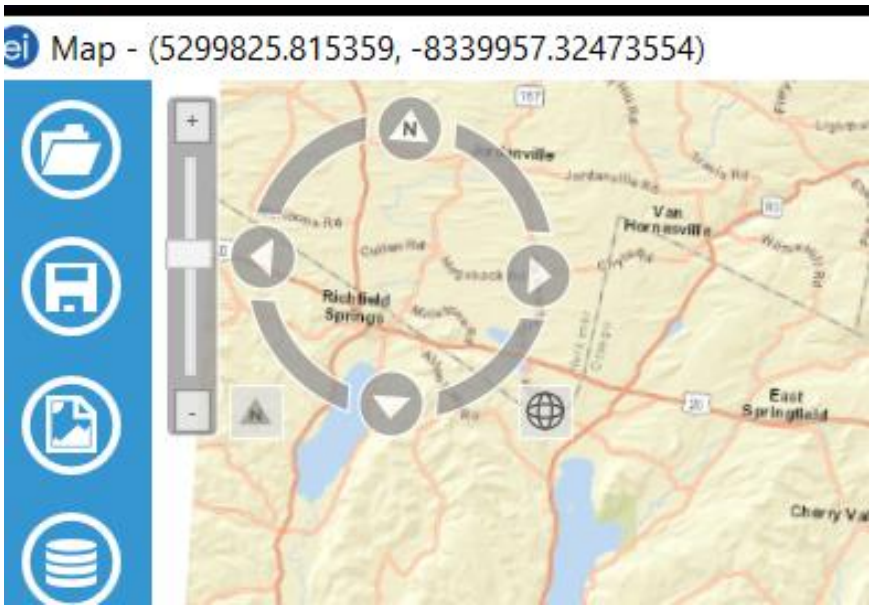
View Map Information / Displaying values

The ALBZCTA region shape file contains census data as part of the associated DBF file. Use your mouse button to hover over different map regions and the value associated with the data file will be displayed (as shown below):



Map coordinates with mouse hover

You can use the Zoom In/Zoom Out control located on the left-hand side of the screen to change the dimensions of the map.

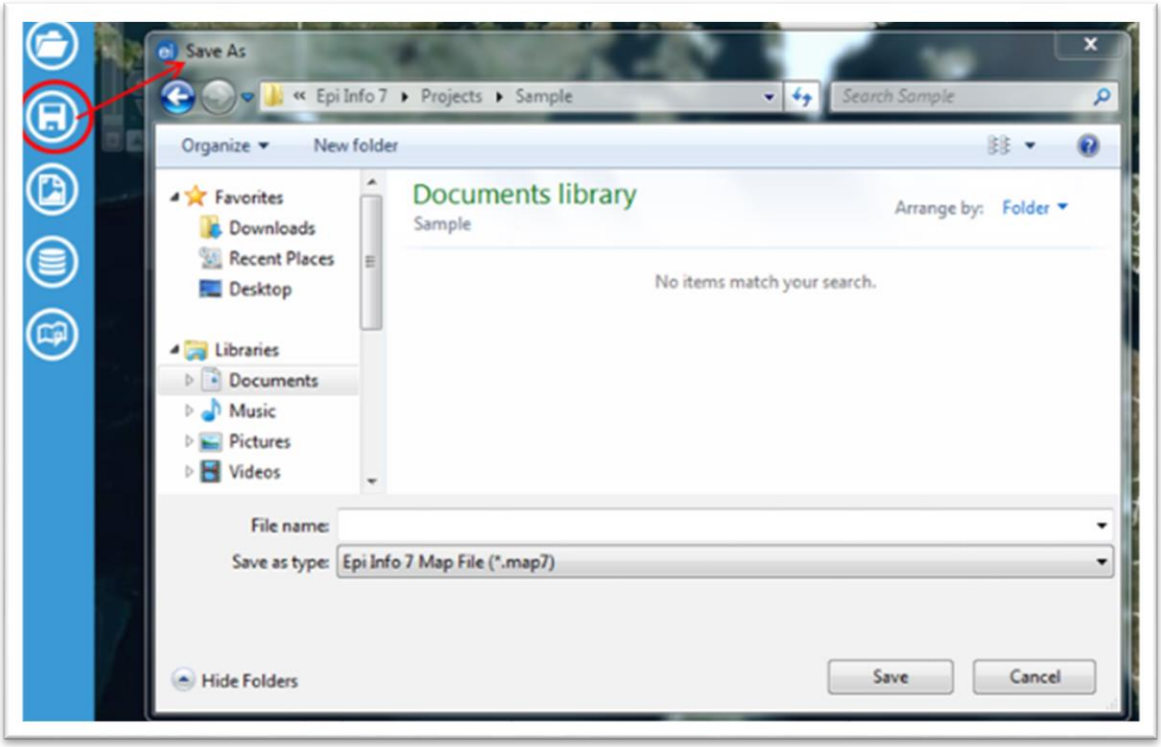


Zoom slider

Saving a Map File

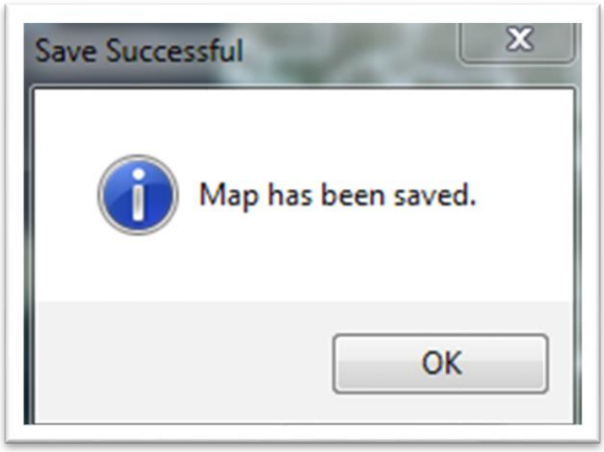
You can save your map in two different ways. The first way saves it as an Epi Info™ 7 Map File MAP7, which keeps all your map's data layers and settings. This type of file only works with Epi Info™ 7, but it has a special feature: when you open it later, it automatically updates if your data has changed. The second way saves your map as an image file PNG. You can open this type of file with any program that displas pictures, like Microsoft Paint. However, image files just save how your map looks - they don't keep any of the data or settings.

To save it as an Epi Info Map file, click on the floppy disk icon (Save) located on the Epi Map menu. When you click Save, a window opens asking you where you want to save your file and what to name it.



Save button and Save As dialog box

1. Choose a file destination from the folder drop-down list at the top of the dialog box and type the desired file name in the **File name** box. The map is saved in the *.map7 file format, which can be opened later by Epi Info™.
2. Click the **Save** button. A **Save Successful** dialog box appears.



Save was successful dialog box

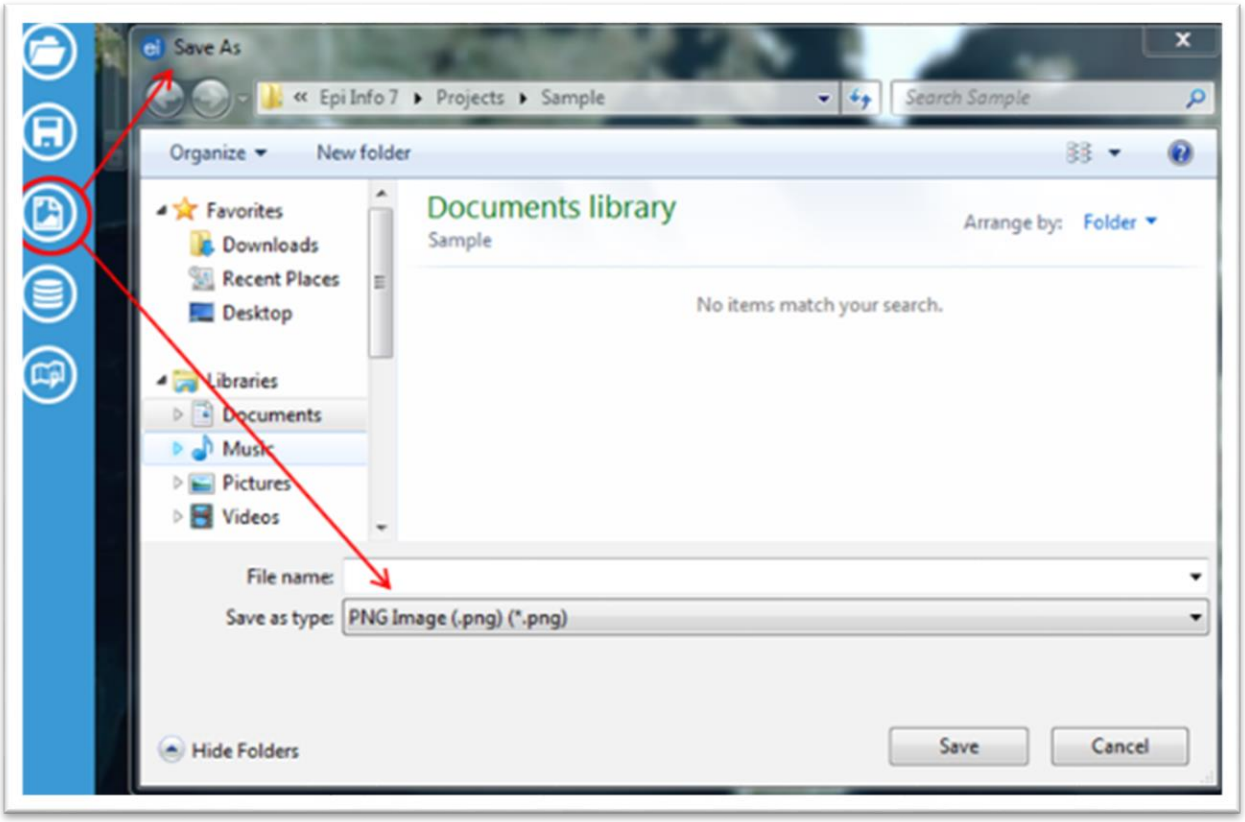
3. Click **OK** to exit.

Save Your Map as an Image

When you save your map as a picture, it *create*s a PNG file that shows exactly how your map looks. You can put this picture file into documents or open it with other programs. To create a picture of your map, click **Save as Image** in the toolbar.

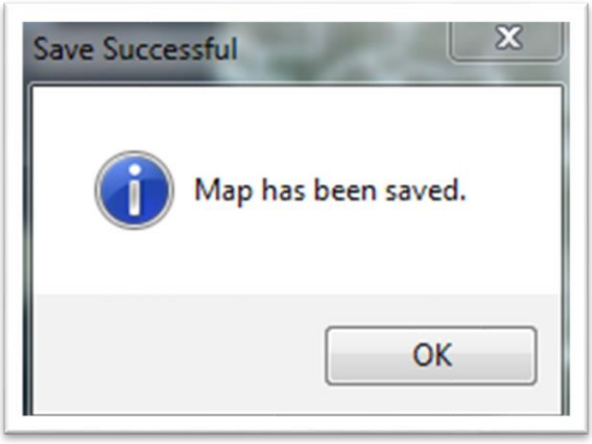
Note: Remember that if you save your map as a picture, you won't be able to open or change it later in Epi Info™.

1. Click Save as Image. A window will open asking you where you want to save your image.



Save As image button and Save As (*png) dialog box

2. Choose a file destination from the folder drop-down list at the top of the dialog box and type the desired file name in the **File name** box. The map is saved in the PNG format, which can be opened later with an image software application.
3. Click the **Save** button. A Save Successful dialog box appears



Save As image button

4. Click **OK** to exit**.**

Creating a Case Cluster Map

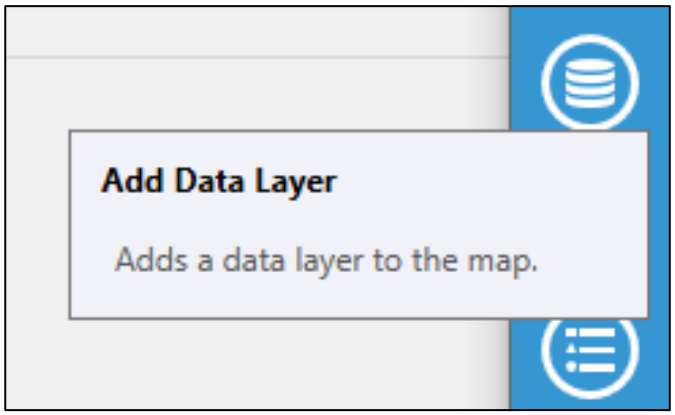
Case cluster maps can be made if you have geographic data for the longitude and latitude measurements of your map corresponding to the region. Latitude and longitude coordinates are developed with GIS tracking information.

You will incorporate a case cluster layer to the existing map corresponding to the location of the households of the children interviewed. You will then incorporate an additional layer to map, as a spot map, for the hospital and school locations in the region. You will use different symbols to show these categories on the map.

This will help illustrate which zip codes had more cases of asthma and the median income corresponding to those zip codes.

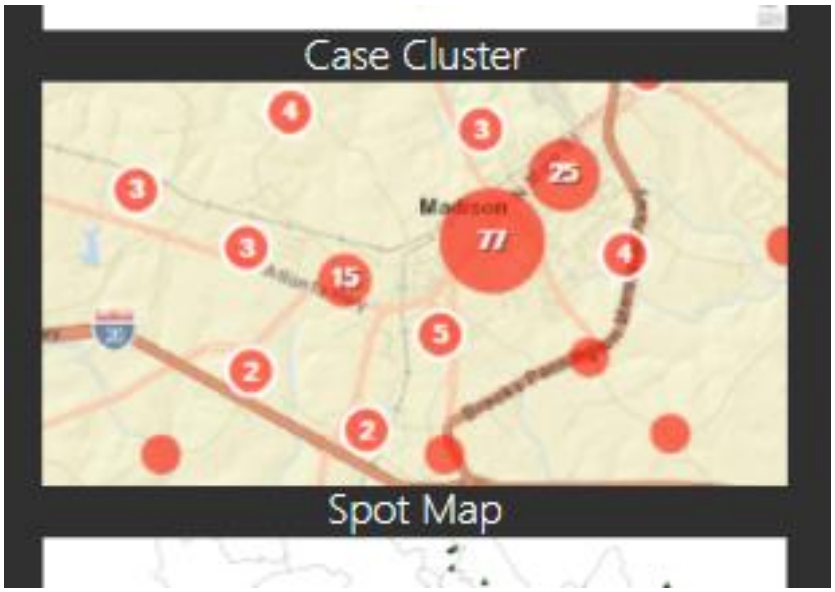
First, we will incorporate the location of the households.

1. From the Epi Map menu, click on the **Add Data Layer** option



Add Data layer

2. Click on Case Cluster.



Case Cluster Spot map

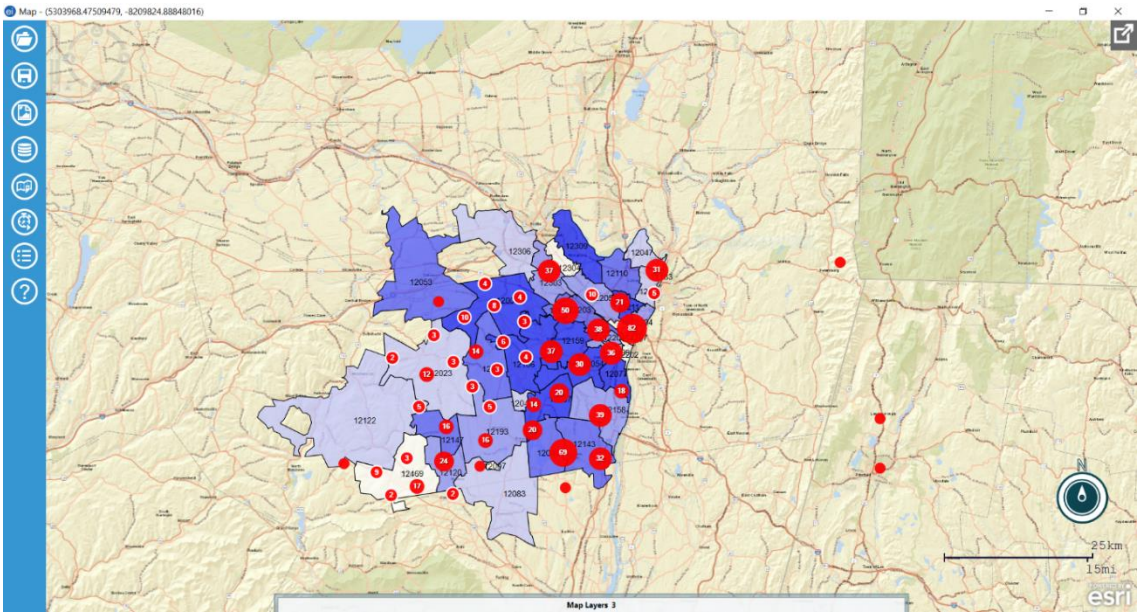
3. Click on the **Browse** button.
4. Select Epi Info 7™ as the **Database Type**.
5. Click **Browse** and navigate to the Community Health Assessment.prj file located in the Epi Info 7/Projects/CommunityHealthAssessment folder.
6. Click **Open**
7. Select the **PreInterventionSurvey** form.
8. Click **OK**.
9. The Latitude and Longitude window opens. Select Latitude from the **Latitude** drop down list.
10. Select Longitude from the **Longitude** drop down list.
11. Type Cases in the **Legend Description** box.

12. Change the Color Representation parameter to Red.
13. Click **OK**.
14. Click **OK**.

Epi Map will immediately display the household locations on the map. The locations are represented with red dots.

- Notice that large clusters of cases appear as bigger circles with the total case count contained inside of them.
- Notice that an additional layer has been added to the workspace.
- You can switch between street and satellite views using the selector located in the top-right corner of the map window.
- The case cluster map was placed on top of the income choropleth. Does the map indicate a correlation between income and asthma in the survey data?

24.



Case clusters

Adding more layers /Spot Map to display Hospitals

Using latitude and longitude measurements, we will now incorporate an additional layer to represent hospitals in the area. This would allow us to demonstrate if there is a lack of healthcare facilities in the region. This layer would be incorporated as a Spot Map.

Select the Add Data Layer option from the Epi Map menu.

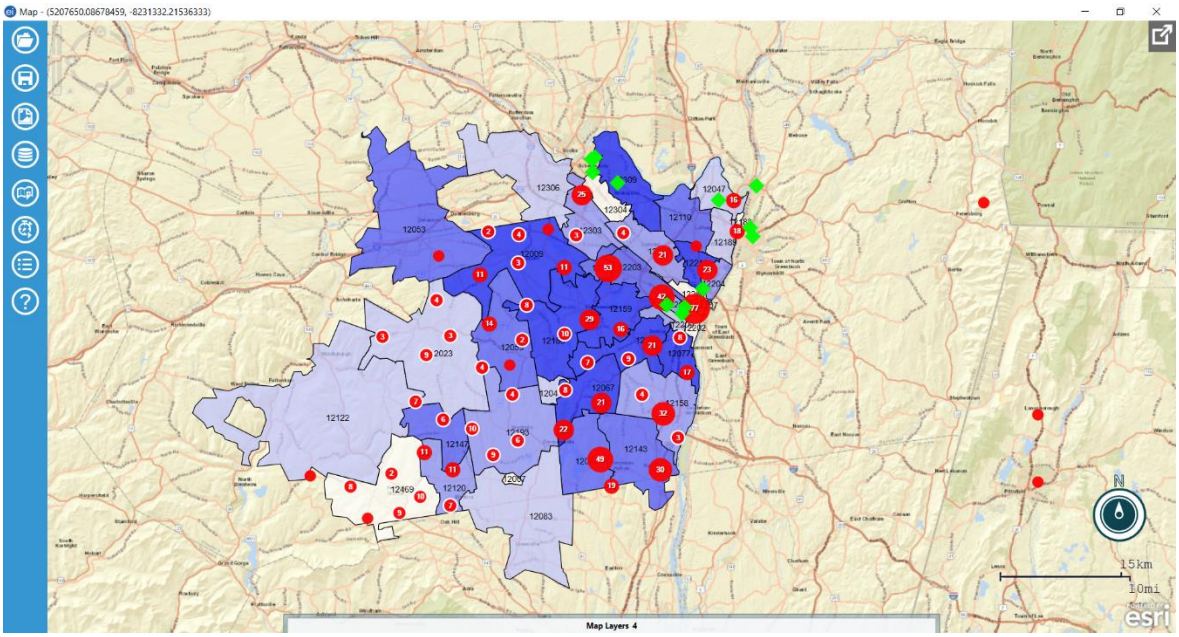
1. Select the **Add Data Layer** option from the Epi Map menu.
2. Click on **Spot Map**.
3. Click on **Browse** button.

4. Select MS Access 2000-2003 (.mdb) as the Database Type.
5. Click **Browse** twice.

6. Locate the file CommunityHealthAssessment.mdb in the Epi Info 7/Projects/Community Health Assessment folder.

7. Click **Open**.
8. Click **OK**.
9. Select the **HospitalPoints** table.
10. Click **OK**.
11. The Latitude and Longitude selection window opens.
12. Select **LAT** from the Latitude drop down list.
13. Select **LON** from the Longitude drop down list.
14. Type **Hospitals** in the Legend Description box.
15. Change the Color Representation parameter to **Green**.
16. Select Diamond from the **Select Shape:** drop down list.
17. Click **OK**.

This map will immediately be displayed incorporating the new layer. Your map might look like this:

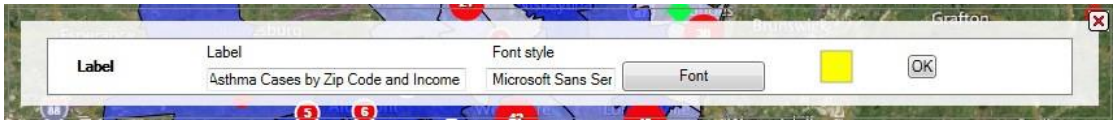


Map view after selections

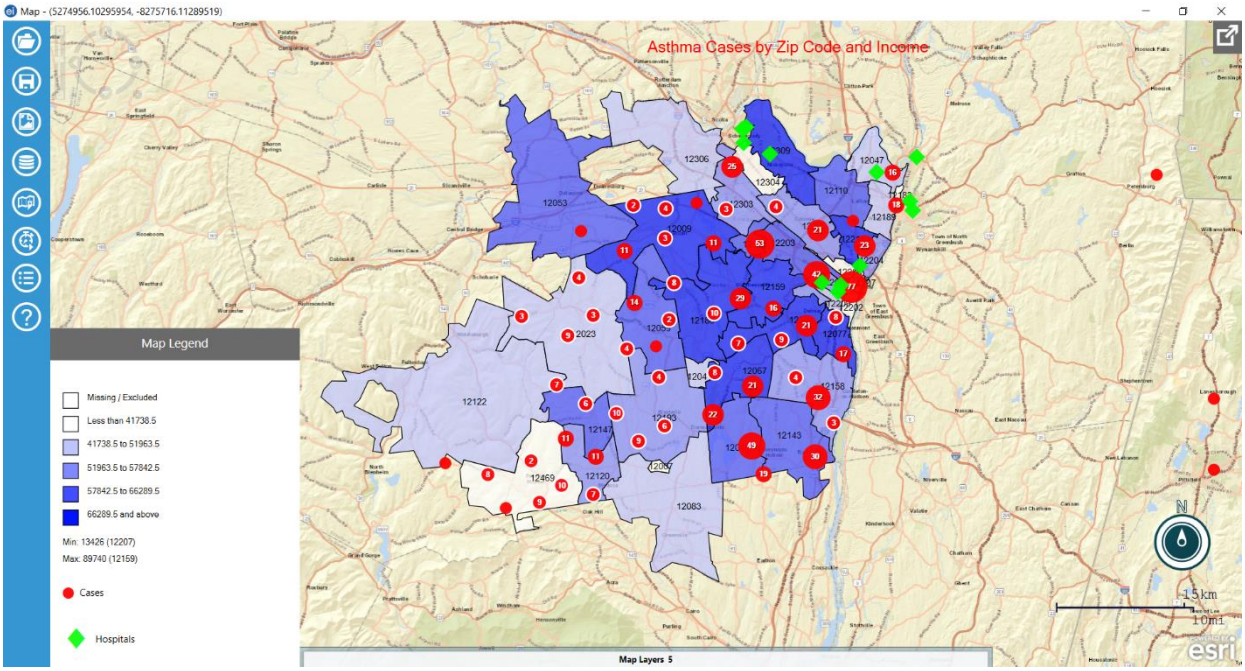
- When you save the points as a new layer, they will appear in the Map Manager as an editable layer.
- In the list of layers, the hospital layer must be the top layer. You can move layers up and down using the Map Layer control in the bottom section of the Epi Map workspace. Use the up and down arrows to move layers accordingly.

Adding more layers /Spot Map to display Hospitals

You will create a title for the asthma and income map.

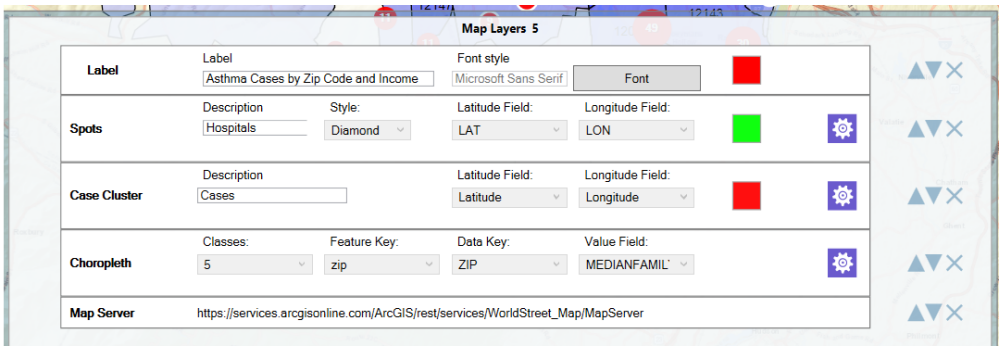
1. From the Epi Map workspace, right click on the exact area where you would like to place a Map Title. A series of options will be available. Select **Add label**. The Add label dialog box opens.
2. In the Label field, **type Asthma Cases by Zip Code and Income**.
25. 

3. Click on **Font** button to modify font settings. Select font type and font size.
4. Click **OK**. The title appears on your map in the exact location where you right clicked.



Map with legend descriptions

- Once the label has been created, a new layer is displayed in the bottom section of the Epi Map workspace. To make any modifications to your label, retrieve the layer and make the desired modifications.
- Please note that the label can't be dragged and moved around the map. You would need to delete the layer and create a new layer by placing the label on the desired location.
- Notice that a new layer has been added to the Map Layer gadget.



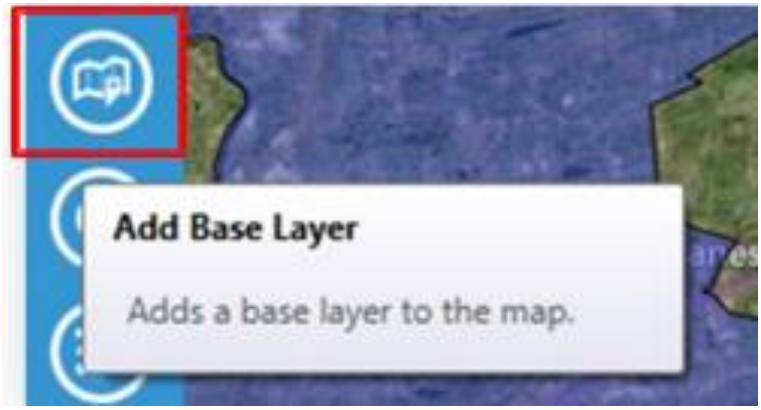
Map layer gadget with gears

5. To remove the label, retrieve the corresponding layer and click on the X.

Adding a Reference Shape Layer

The data and zip codes contained in the survey information crossed county lines. To see how this affects the locations of the schools in the survey, you can place the county shape over the region shape.

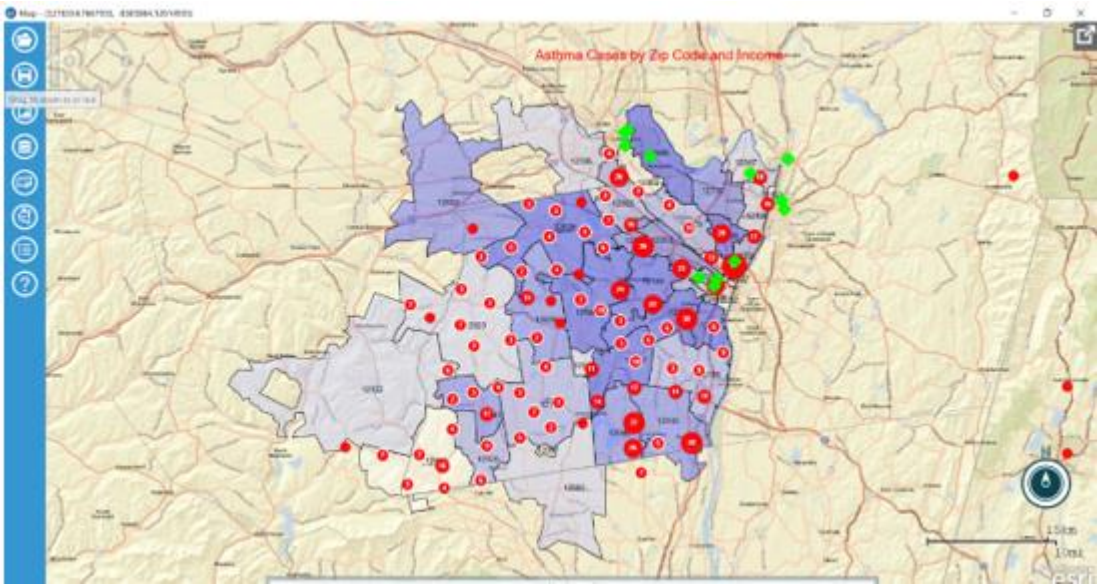
1. Select the **Add Base Layer** from the Epi Map menu.



Add Base Layer

2. Click on the **Shape File** radio button.
3. Click **Browse**.
6. Select From Shape File.
7. Locate the file Albany County Boundary_region.shp in the Epi Info 7/Community Health Assessment folder.
8. Click **Open**.
9. Click **OK**.

The map should look like the following:



Asthma clusters mapped

In the list of layers, move the latest layer three levels down to display the different stratifications properly. You can move layers up and down using the Map Layer control located in the bottom section of the Epi Map workspace. Use the up and down arrows to move layers accordingly.

Saving a Map as an Image

To save a map to be used in presentations or documents, save it as a .png file.

1. Click on the **Save as Image** icon.



Save as an image feature

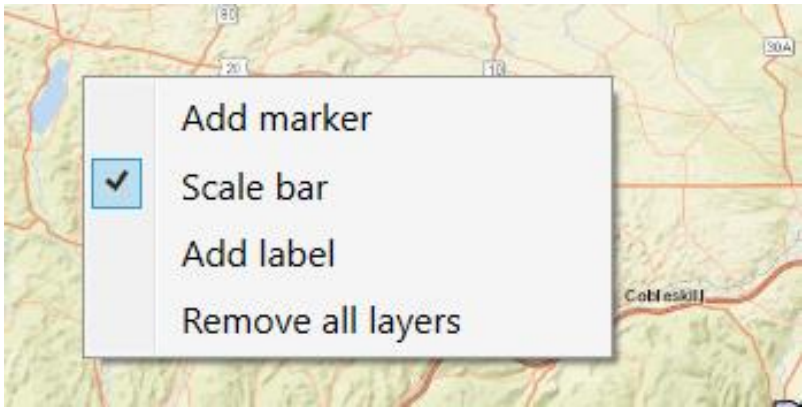
2. Name the file **SchoolPointsMap**.
3. Click **Save**.

Basic Functions

The map window toolbar contains the following tools to help create a meaningful representation of your data:

1. Add marker
2. Scale bar
3. Add label
4. Remove all layers

You can access these basic functions from the main map window by pressing the right click button on your mouse anywhere on the map canvas (called a "right click").

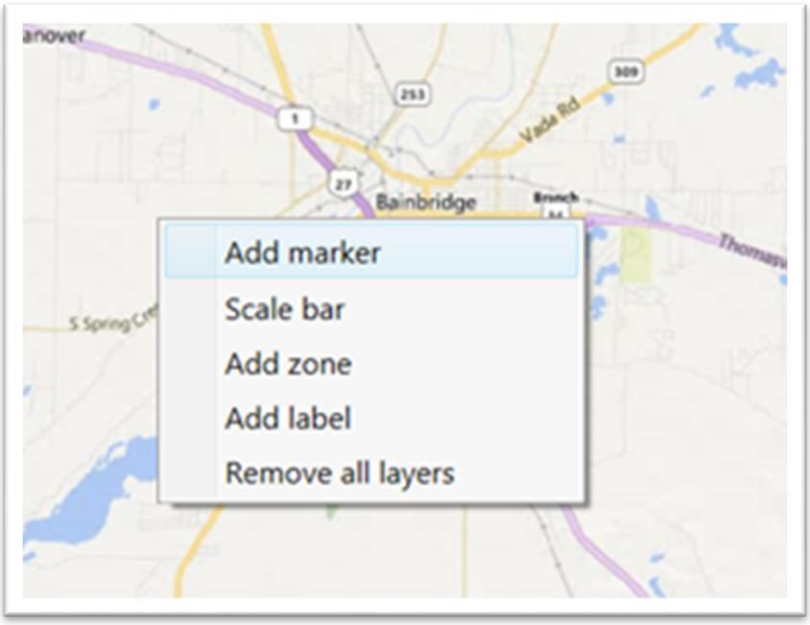


Basic functions menu in Epi Map

Add a Map Marker

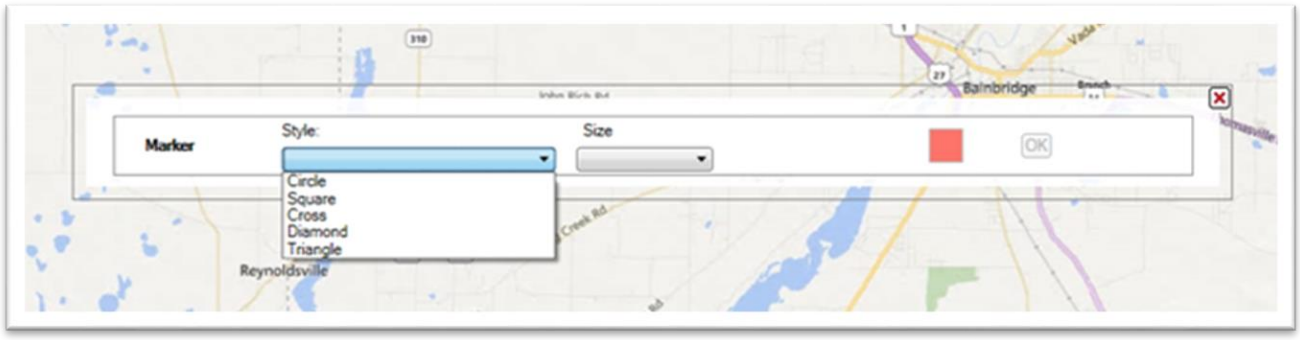
A marker helps you highlight important places on your map by using symbols. Each marker can have a different shape and size to help you tell them apart.

1. Right click on the main map window in the location where you would like to place the marker. A menu appears displaying the available options.
2. Select Add marker.



Basic functions menu showing Add marker

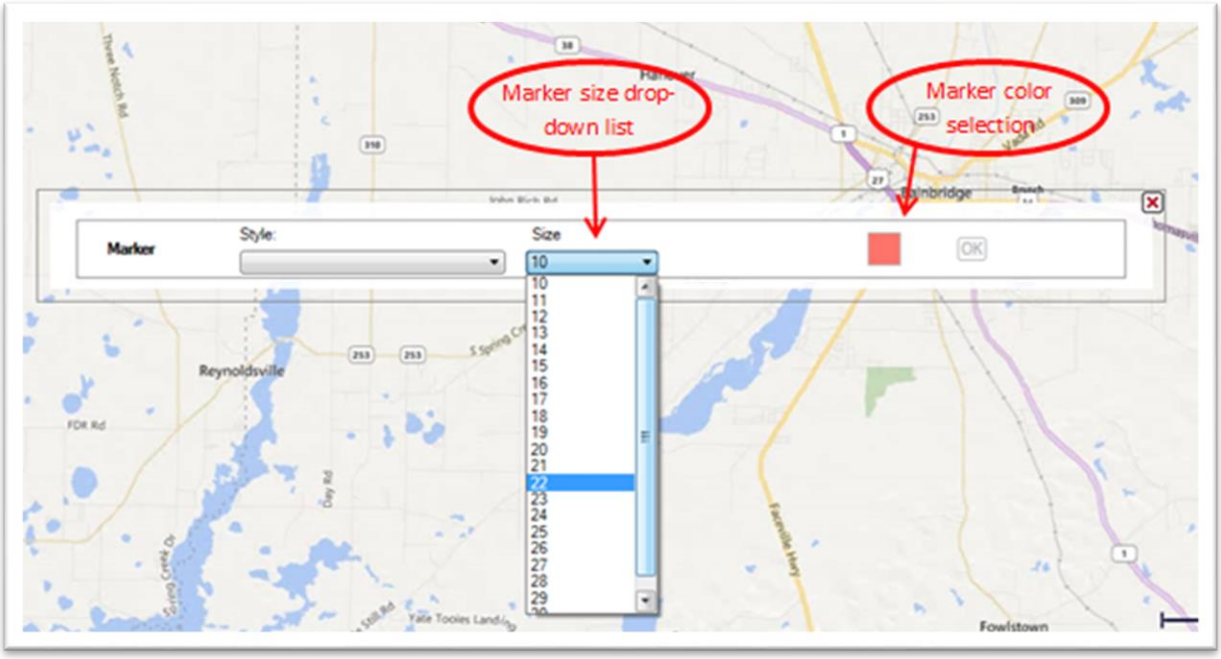
3. Select a **Style** from the Style drop-down list for the marker. The style will determine the marker's shape on the screen.



Add marker drop-down with marker styles

4. Select the **Size** of the marker from the Size drop-down list. The smallest marker size available is 10, while the largest is 30.

5. Select the **Color** of the marker by clicking on the red **Point Color** button to the right of the dialog box.



Marker size drop-down menu and marker color selection option

6. Click **OK**. The example illustrated below is a diamond-shaped marker of size 20 in red.



Marker displayed in map window

Scale Bar

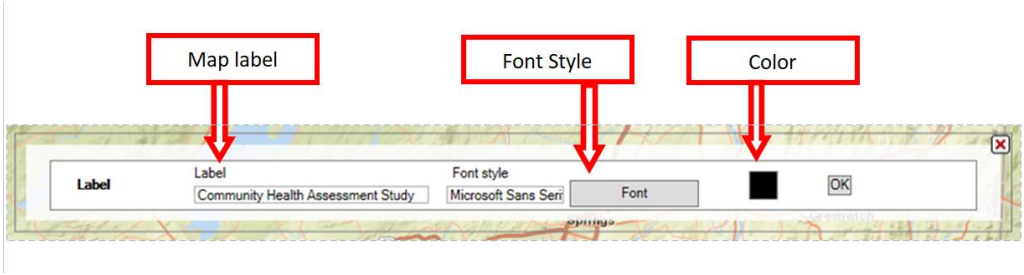
The **scale bar** helps you understand distances on your map. You can show or hide this measuring tool, which appears in the lower right corner of your map window, by using the Scale bar option in the basic functions menu.

Add a Label

Labels help you name important spots on your map. To add a label:

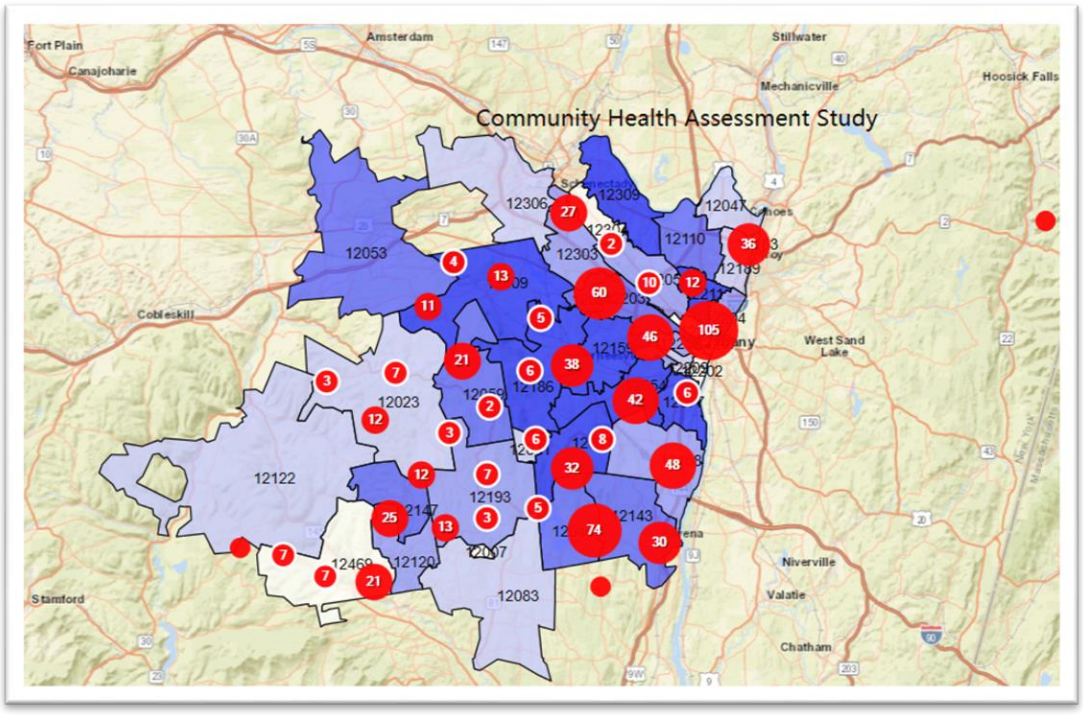
1. Right click where you want the text to appear and select **Add label** from the menu.
2. Type your label's text in the box that appears.
3. Click **t**he **Font** button next to the text box. **You can** make your label easy to read by picking a font from the **font** list.

4. Click the color box and choose any color you want to help your label **stand out**.



Label entry box and font and color selections

5. Click **OK**. In this sample, the label is 16-point Arial font in blue.



Label displayed in map window

Remove Map Layers

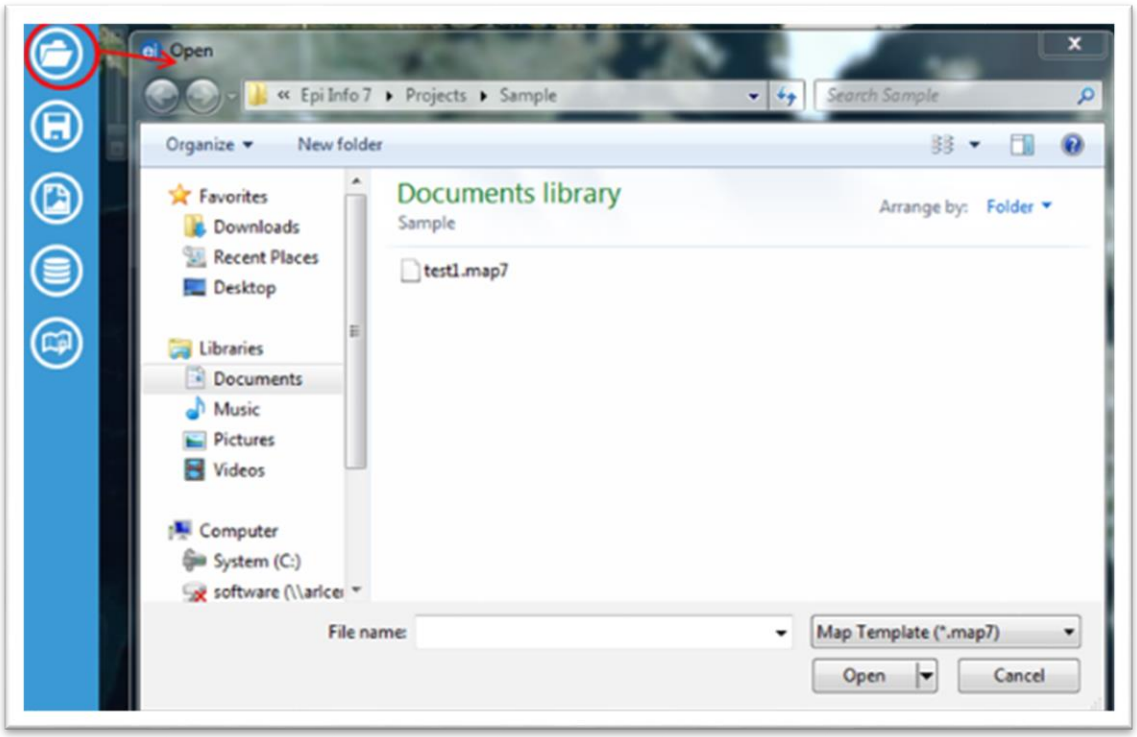
Everything you add to your map - markers, zones, and labels - becomes a layer. When you want to start fresh, you can remove all these layers at once. Right click anywhere on your map and select **Remove all layers.**

Open Your Map Files

Epi Maps works only with Epi Info™ 7 Map files, which have an .map7 file extension. To open one of your saved maps:

1. Click the **Open** button in the toolbar.
2. Find and click the map file you want to use.

3. Click **Open**. The map file updates and displays in the main map window.



Open button and resulting file dialog box

More Maps – Case Cluster

Case Clusters

Case Clusters show where cases are located on your map using geographic coordinates. To make a Case Cluster map, your data must include numbers for latitude and longitude. While the software can turn street addresses into these coordinates (called geocoding), a street address alone won't work unless you use this feature. To learn more about geocoding, check the Form Designer section of this guide.

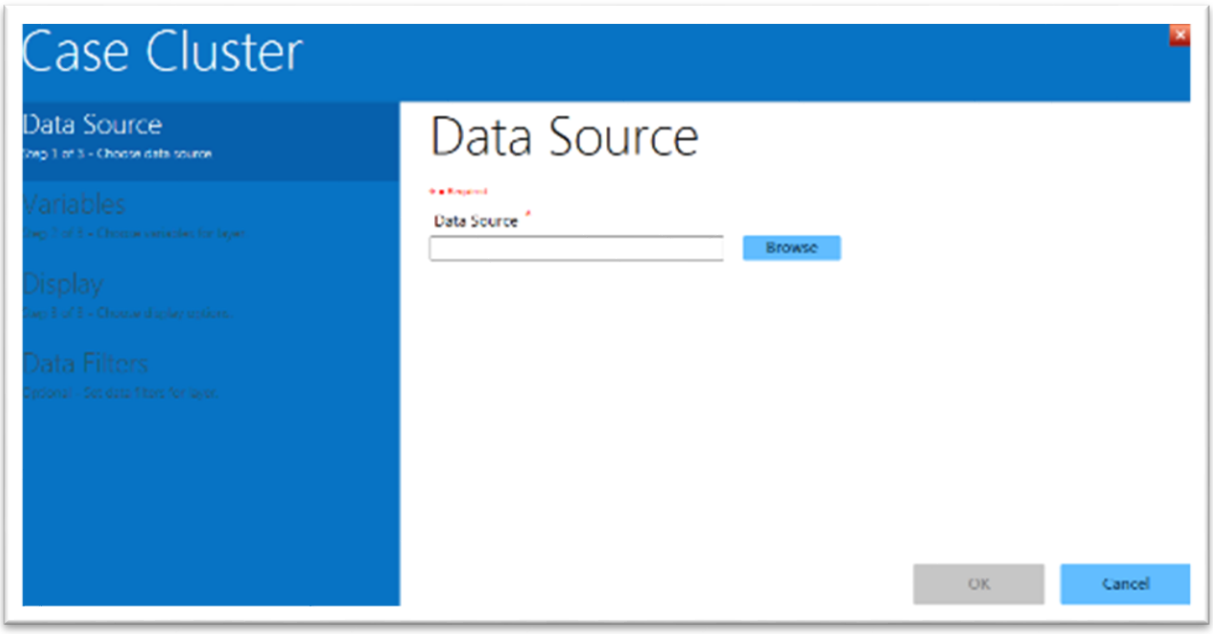
On your map, large groups of cases show up as big circles with a number inside

showing how many cases there are. Single cases appear as dots without numbers. Here's how you view your case clusters:

Larger circles mean more cases are in that spot. When you look closer by zooming in, these big circles break into smaller ones or single dots. If you move your mouse over a small cluster (fewer than 12 cases), the cluster spreads out to show each case separately. This helps you see when multiple cases come from the same place, like one household.

Create Another Case Cluster Map / Using E.Coli data

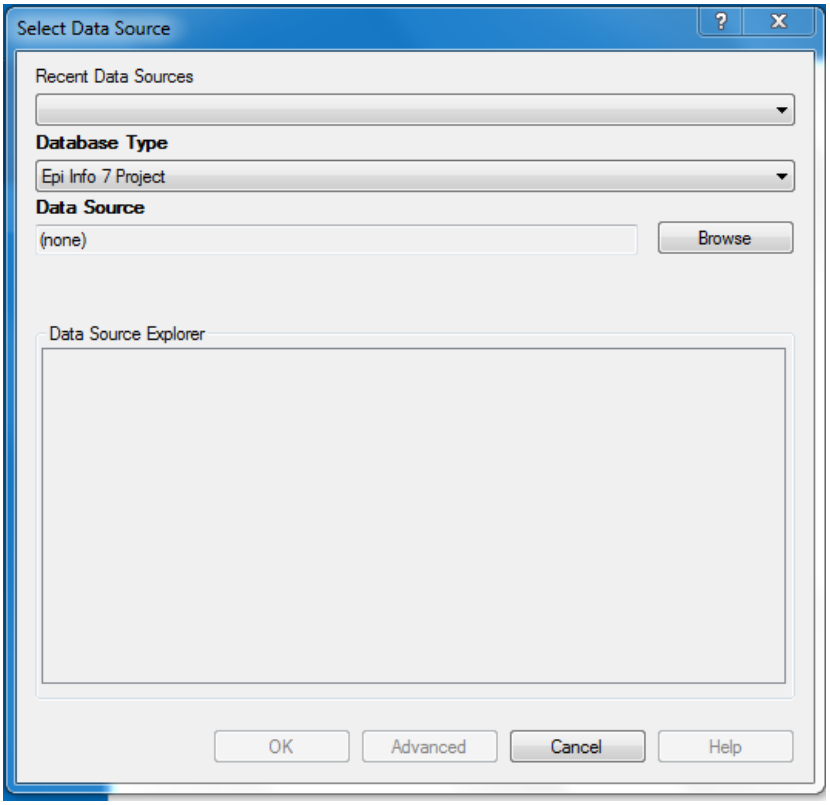
1. Click **Add Data Layer** in the toolbar, then choose **Case Cluster**.
2. Select your data source from the Case Cluster layer dialog box.
3. Click the blue **Browse** button next to the blank Data Source line.



View Data Source dialog box

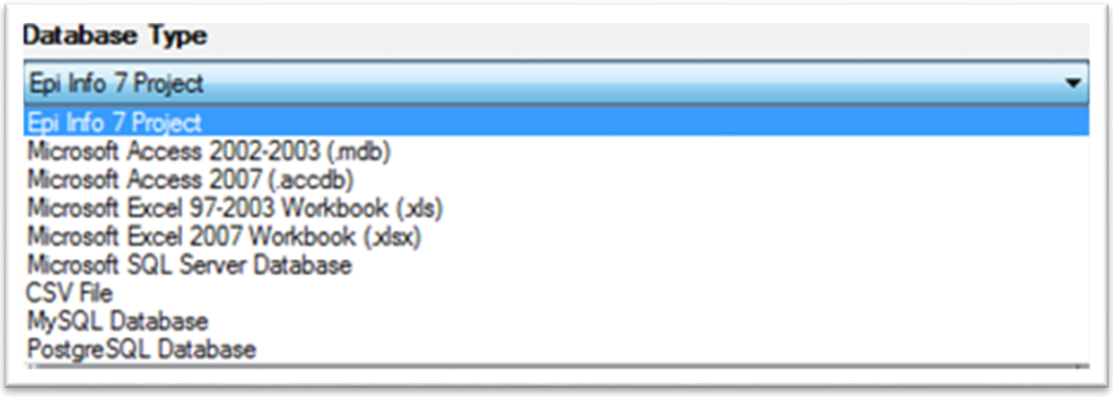
Selecting a Data Source is the first step in the Case Cluster process.

1. A Select Data Source dialog window will open.
2. Select the appropriate database option from the **Database Type** drop-down list.



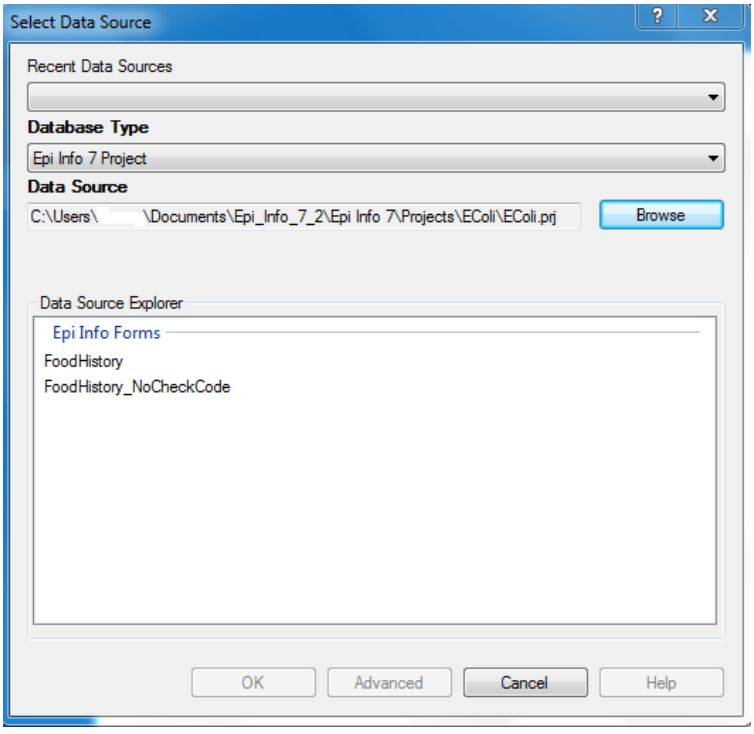
Select Data Source dialog box

In the database window that opens, pick your type of database from the list you've used a database recently, you can find it in Recent Data Sources.



Database Type drop-down menu

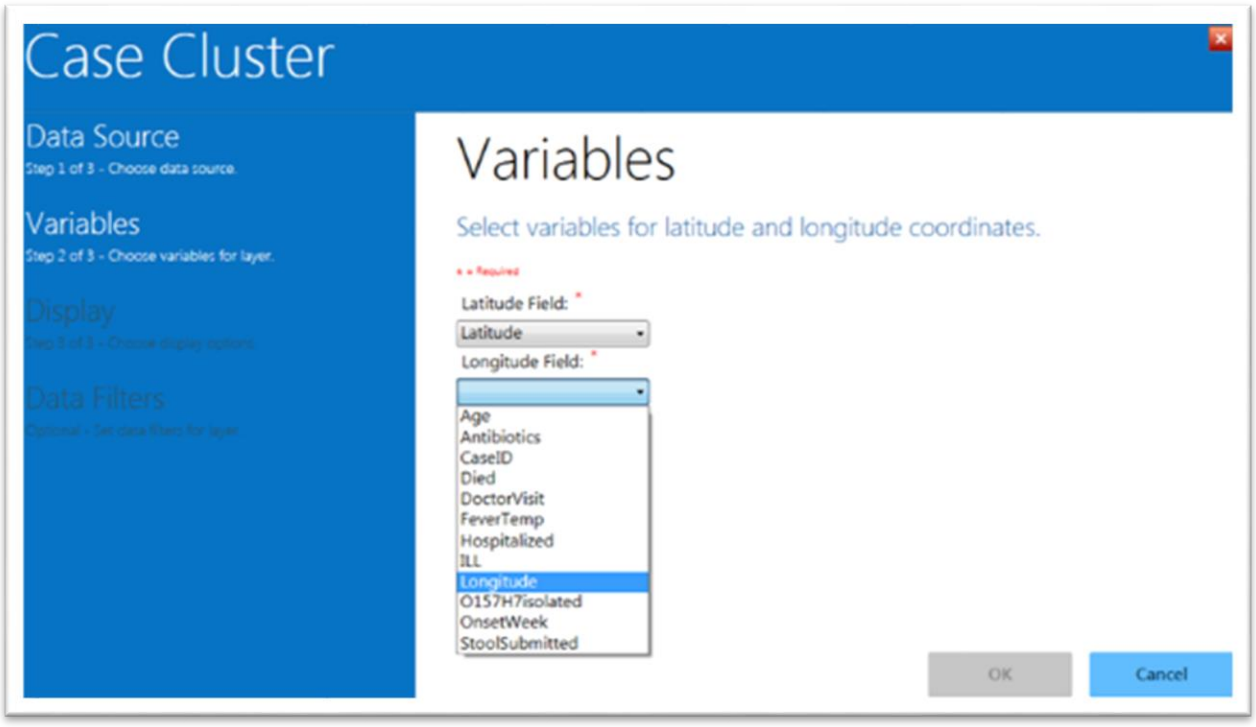
1. Choose your database type, click **Browse** to see your files.
2. Choose your file by double-clicking it. Your file name will show up in the Data Source field, and you'll see your project data.



Project Data

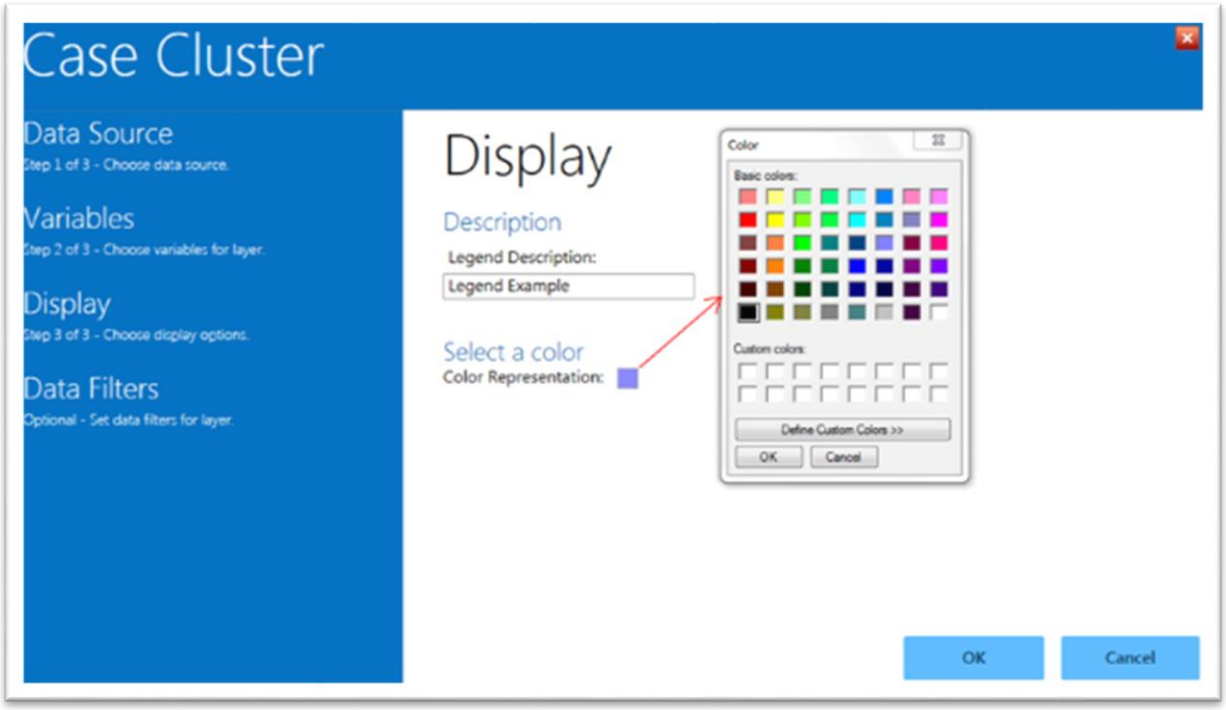
1. Select the data you want to map and click **OK**.

2. Choose **Latitude** and **Longitude** from their drop-down lists.



Case Cluster Variables list

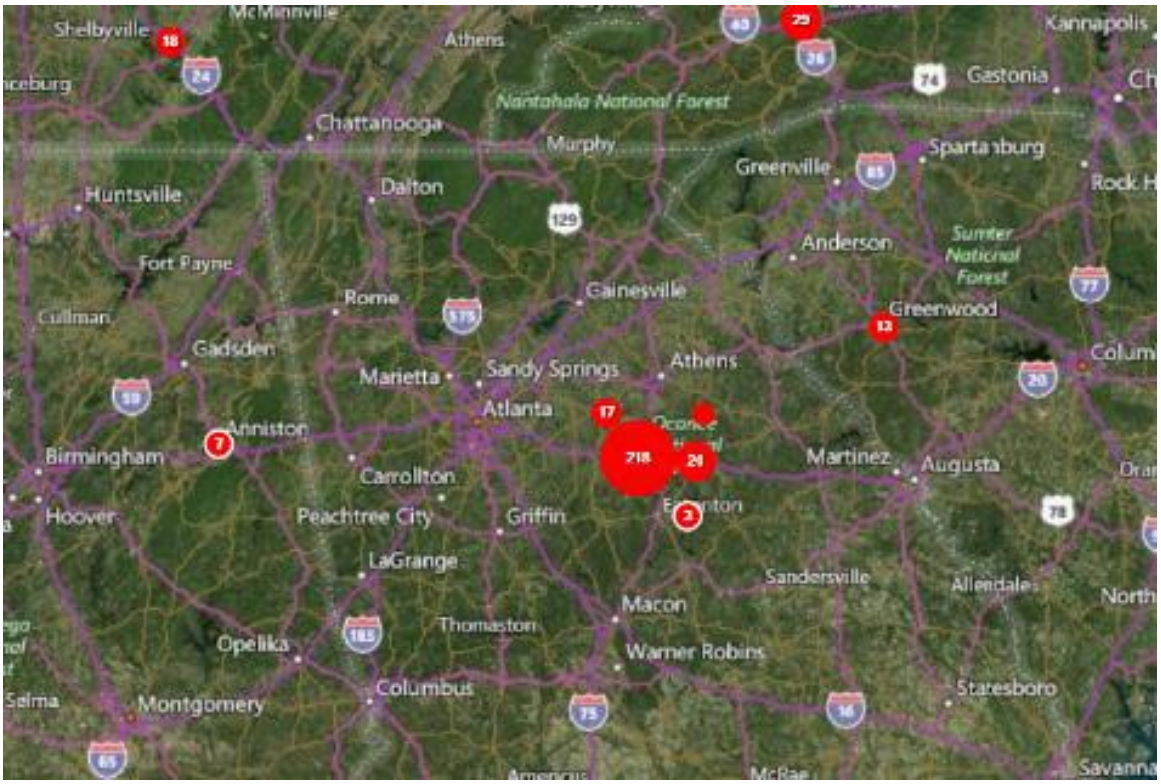
Now you can make your map look how you want. Give your map a name in the Legend Description box. Pick a color for your clusters by clicking the color box - it starts as blue, but you can change it to any color you like.



Display options

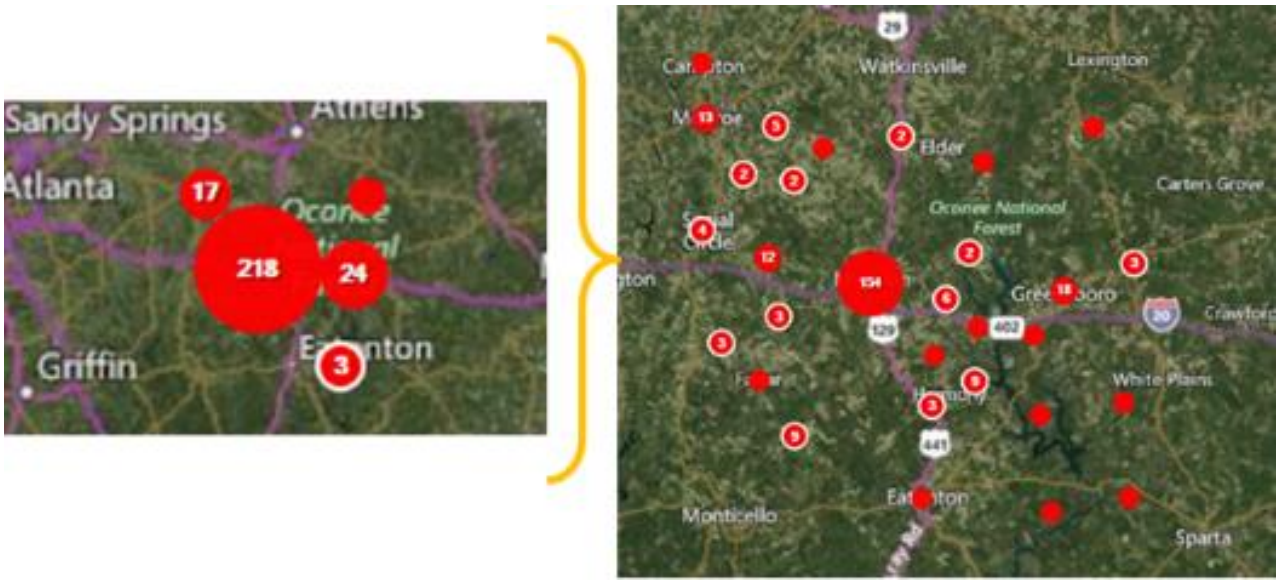
Begin your Case Cluster Display options:

26. Click **OK**.
27. The Case Cluster layer closes and returns to the map. Red circle "clusters" appear in the appropriate locations, with the number of cases in the center of each circle.
28.



Case Cluster results

29. As the view zooms in, case clusters separate into smaller clusters and individual cases appear as single red dots.



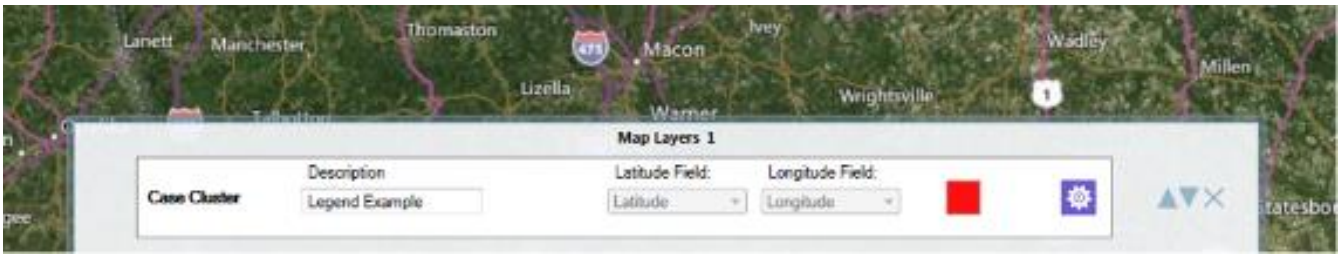
Case Cluster zoom

30. Moving the mouse over a small case cluster (less than 12 records) will "flare" the cluster. The number seven at the center of the cluster in this example represents seven different cases. When the mouse is over the case cluster, seven smaller dots appear, each representing one of the records inside that cluster. This is useful if the same household contains multiple cases.



Expanded case cluster

31. Hovering over the **Map Layers** bar at the bottom of the map will scroll the bar open and display current map layers (e.g., **Legend Description**, fields).



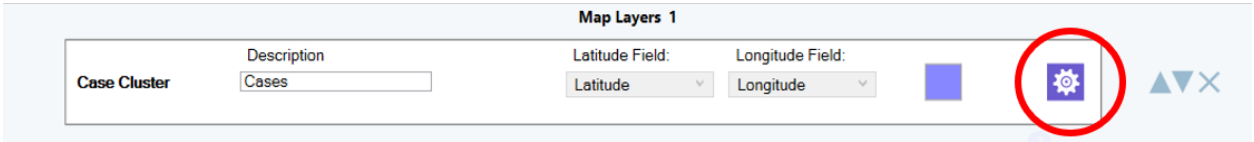
Map Layers bar at bottom of map window

Data Filtering in Epi Map

Data filters in Epi Map are used to select a subset of data by specifying and applying certain conditions. This allows the user to show the effect of a variable on the geographic distribution of cases. To access the data filter tool, move your cursor over the **Map Layers** bar at the bottom of the main map window as described in the previous section. In this example, there is currently one case cluster layer, which is red and does not contain any filters.

The following example demonstrates how to apply a filter to the **Age** variable in the data layer from the previous example.

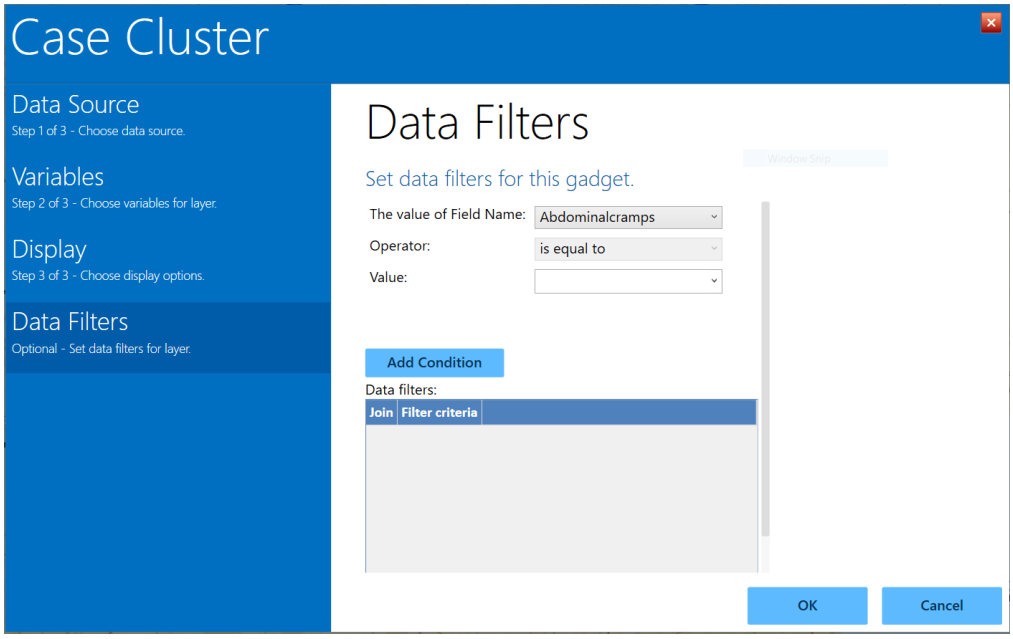
1. In the **Map Layers** bar, click the **Edit data layer** tool.



Edit data layer icon in map window

1. In the **Map Layers** bar, click the **Edit data layer** tool.

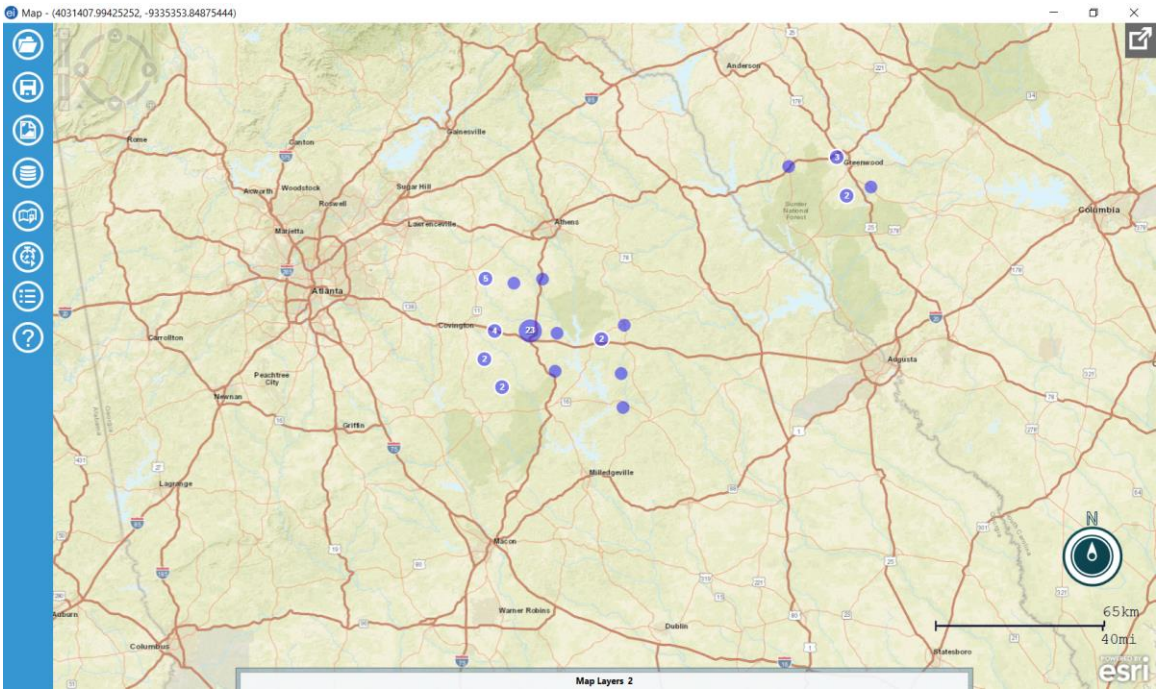
2. The **Case Cluster** layer opens. Select **Data Filters** from the list of options in the left-hand menu.



Data Filters menu in the Case Cluster layer

3. From the drop-down list labeled **The value of Field Name**, select **Age**.
4. From the **Operator** drop-down list, select **is less than**.
5. Enter **21** into the **Value** textbox.

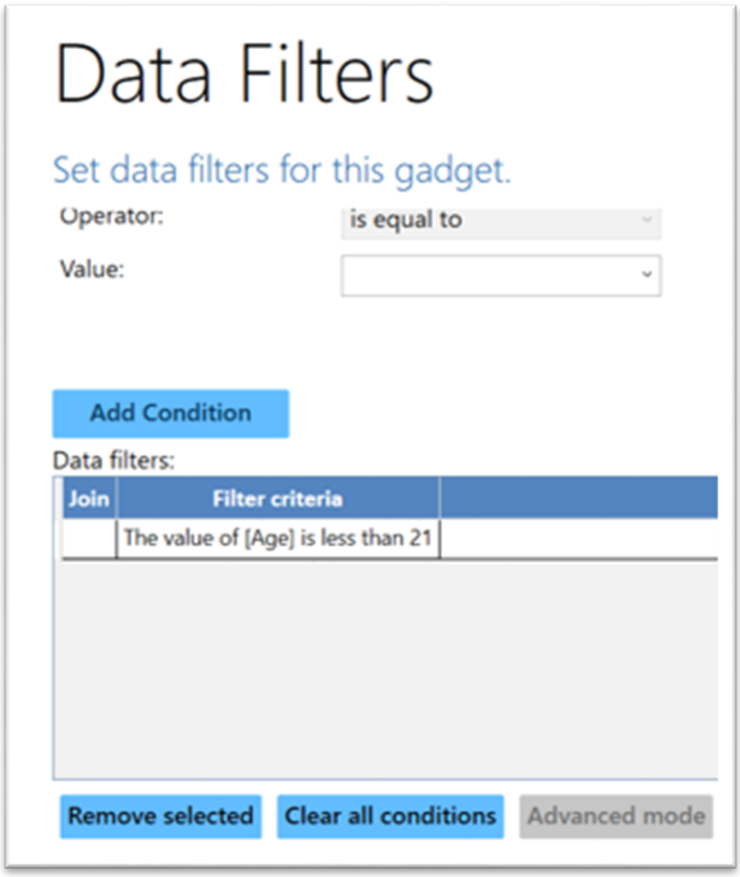
6. Click **Add Condition**. One row defining this filter is automatically added to the **Data filter** table.



Data filters menu with one filter added (age less than 21 years)

7. Click **OK**. The map refreshes to display only cases in which the patient's age is less than 21 years.

    a) A filter can be removed by scrolling down below the **Data filters** window, highlighting the desired filter, and clicking **Remove selected**. Clicking **Clear all conditions** removes all filters.



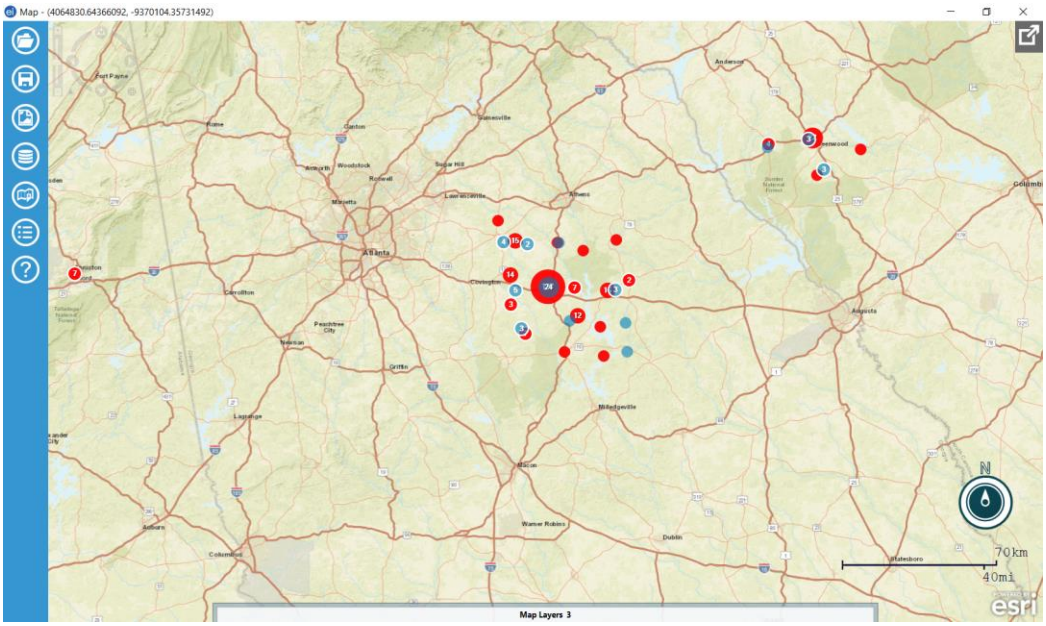
Remove one selected or all data filters

Visualizing Multiple Data Layers (Additional Data Layers)

Filters are a powerful tool for sorting data. In certain instances, however, it is best to display the full dataset in groupings displayed in different layers. This requires separating the data, by filtering on more one or more conditions for each layer. After completing the steps to this point in the user guide, the example map displays the patient population that is less than 21 years of age. To create a map with one or more layers, another data layer is needed to show an additional age group in the patient population. In the following example, this is achieved by

adding another case cluster data layer with a filter based on the patient's age. Blue clusters will represent the records that meet the specified conditions.

1. Select **Add Data Layer** from the toolbar.
2. Select **Case Cluster** from the drop-down list.
3. Follow the **Data Source** selection steps from the previous **Case Cluster** section to select the **Ecoli.prj** folder and the **Food History** form in the **Data Source Explorer.**
4. Select **Latitude** from the Latitude drop-down list.
5. Select **Longitude** from the Longitude drop-down list.
6. Enter **Legend Example2** in the **Legend Description** box. Leave the color as the default blue.
7. Click the **Data Filters** option in the left-hand menu.
8. The Data Filters menu appears. In the drop-down for **The value of Field Name is**, select **Age**.
9. From the drop-down list, select **is greater than or equal to**.
10. Enter **21** into the **Value** textbox.
11. Click Add Condition.
12. Click **OK.**
13. The map window displays both age group data filters by color.



Example of case clusters filtered by two age groups

The blue clusters represent cases in which the patient is 21 years of age or older, while the red clusters represent cases in which the patient is less than 21. **Epi Map** tool allows for multiple age groupings and can also filter the dataset based on other variables (e.g., whether the patient is male or female, foods eaten, etc.).

Note that the Map Layers bar at the bottom of the screen displays a (2) instead of a (1). This indicates that there are two different layers with corresponding filters. The properties of each layer can be viewed by clicking the purple gear icon on the right-hand side (this opens the full **Case Cluster** menu). The order of layers can be changed using the gray arrows, and layers can be removed fully by clicking the x" at the far right of the bar.

Time Lapse

When a case cluster layer is added to Epi Map, the tool displays all cases contained in the dataset (or, if a filter is applied, all cases that meet the selected filter criteria). The Time Lapse tool creates a dynamic environment that illustrates how the dataset transforms over time. To enable this feature, use a dataset with a time variable.

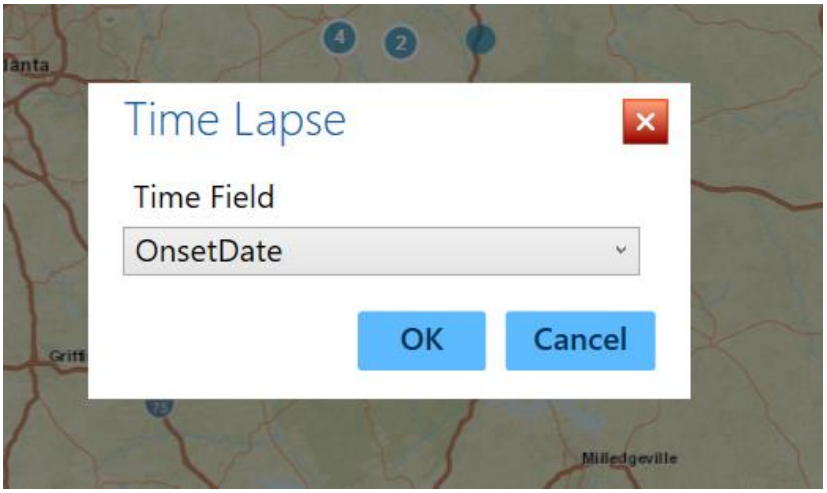
1. Add a **Case Cluster** data layer (see **Adding a Data Layer**). Use the Food History form in the E. coli project for the following example. Select **Latitude** as the latitude variable and **Longitude** as the longitude variable in step 2 of the data layer process.
2. Once the data layer is added, the **Create Time Lapse** icon will appear in the left-hand toolbar**.** Click the icon.



Create Time Lapse icon on toolbar

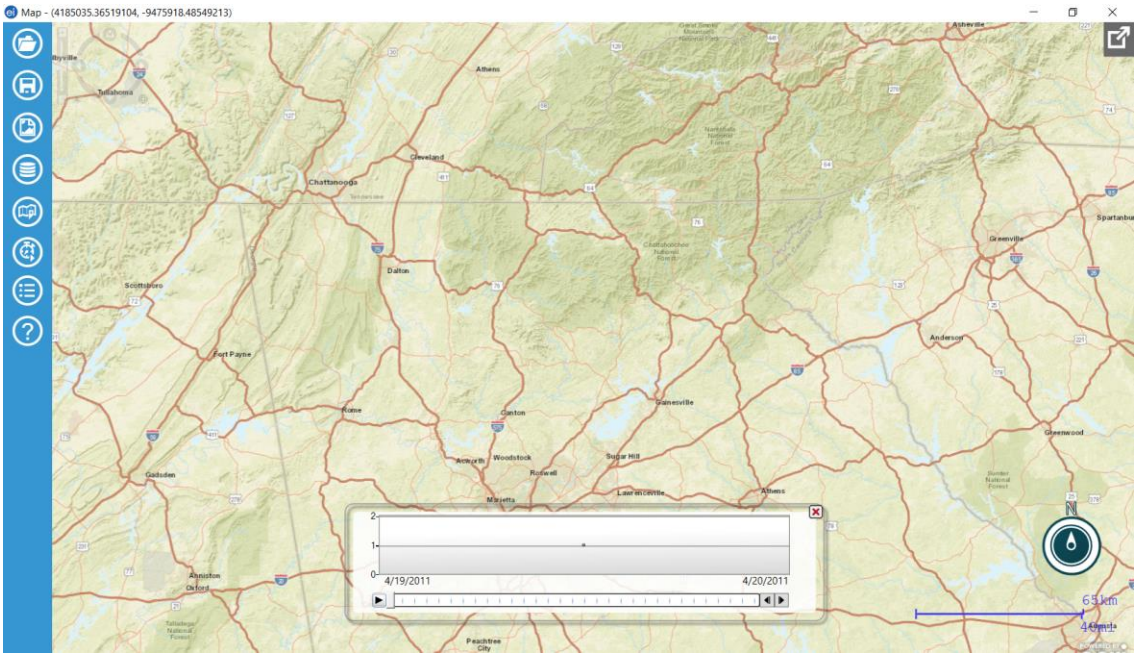
3. Select the **OnsetDate** from the **Time Variable** from the drop-down list. Click **OK**.

32.



Configure Time Lapse dialog box

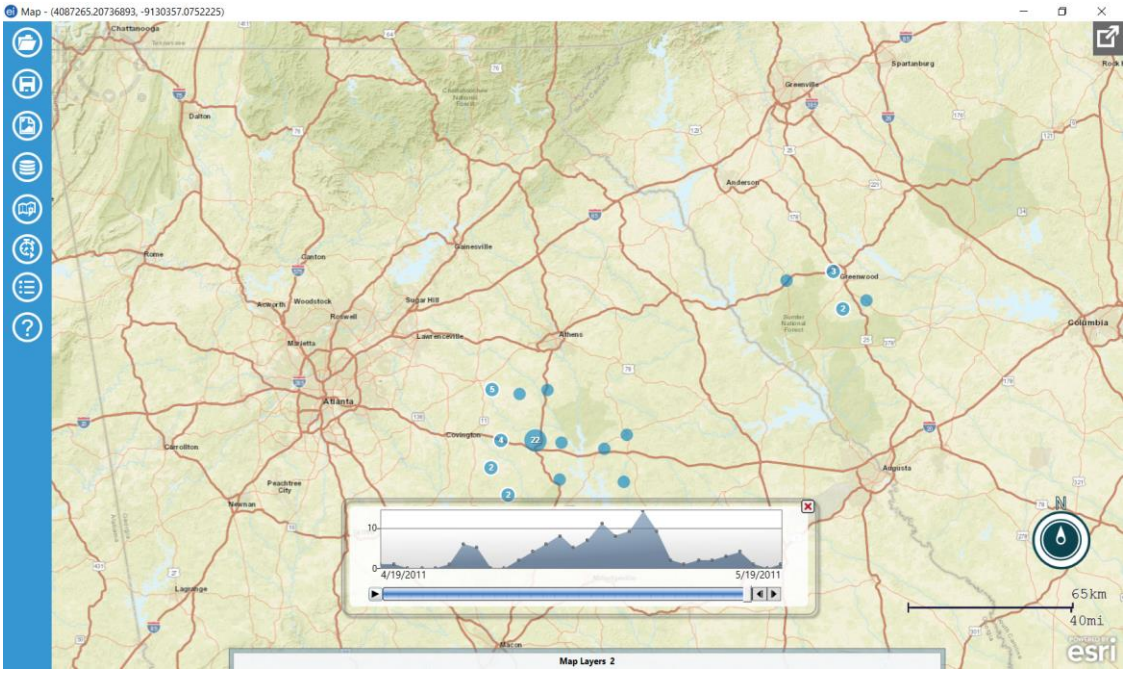
4. The system will add a timeline to the map illustrating the start and end dates found in the dataset. In this example, data in the dataset starts (first onset date) on 4/19/2011 and ends 4/20/2011 (last onset date).
5. To watch the time lapse progression of the symptom onset date over time, click the **Run** button (single right-pointing arrow at the far left of the timeline).



Time Lapse image

The main map window clears and begins adding the cumulative number of cases respective to the timeline. The figure below displays the number of cases in the dataset

from 4/19/2011 to 5/13/2011 (end date is displayed at the lower right of the timeline).



Pause and forward/backward buttons

6. The distribution of cases displays at the bottom of the screen.

Creating Choropleth and Dot Density Maps with Boundaries

Epi Map can create a choropleth or dot density map by combining a dataset with three boundary formats: shapefile, map server, or KML (Keyhole Markup Language) (descriptions of each format follow). This is the map/boundary format compatibility:

- Shapefile – an Esri vector data format for location, shape, and geographic attributes

- Map server – an Open-Source repository for map images and vector data
- KML – an XML notation for geographic annotation and visualization

The boundaries are independent of the dataset but are joined with database keys.

Note: The dataset and boundary file must contain the proper database keys, and the user must designate the proper key from each drop-down list to create a choropleth or dot density map.

Key descriptions are as follows:

- **Feature Key**—designates the variable in the boundary set that will match a corresponding variable in the dataset.
- **Data Key**—designates the variable in the dataset that will match a corresponding variable in the boundary set.
- **Value Field**—designates the value being displayed.

Shapefiles

A shapefile stores non-topological geometric and spatial information in vector format. These files are simple to use but lack complex data elements. Various sources attach additional information or tables to shapefiles for more advanced analysis.

KML

KML is an open-source specification for describing geographic data. Like shapefiles, KML files contain instructions used by mapping tools to draw

boundaries, points, and other feature sets. A benefit to using KML files is that they can be edited using simple text editors.

Map Servers

Map server is a platform used to publish spatial and geographic data to the Internet. One advantage of the map server format is the capability of creating a central repository for mapping data with relational database management systems.

Choropleth Map

A choropleth map uses graded differences in shading or color to display variations of a variable across a geographic area. The color gradient typically spans from one color to another or from a lighter to darker shade of the single color.

The following example demonstrates how to combine a map server feature set for the state of Maryland with the Lyme data set included with Epi Info to create a choropleth map. The boundary format in this case is KML.

Note: This example uses case-based data, where each row in the data set represents an individual case, rather than aggregate data.

---

1. Select **Add Data Layer** from the toolbar.

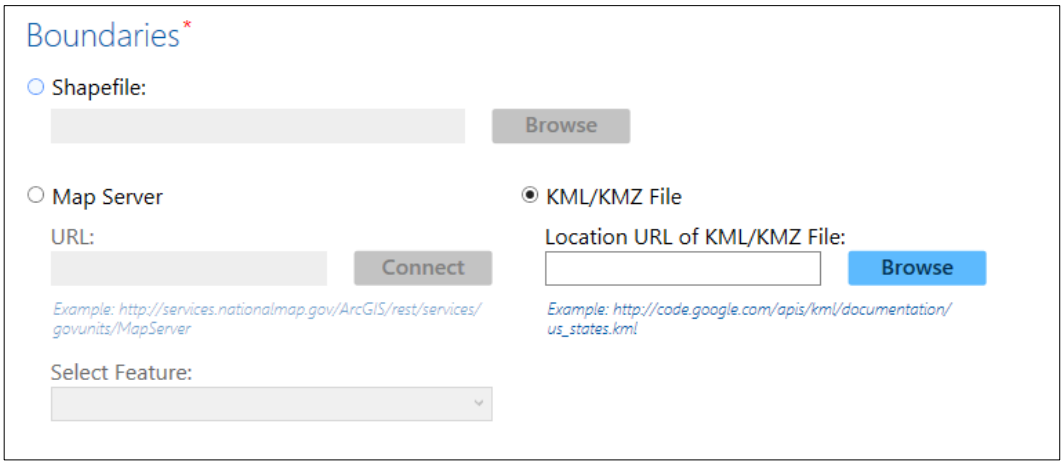
2. Select **Choropleth** from the list.

   The Data Source dialog box appears. Click **Browse** to open the database options.

3. Click **Browse** next to the Data Source field. Select the Lyme folder and then the Lyme.prj file.

4. Select the **Case Report** form in the Data Source Explorer. Click **OK**.

5. Select the KML/KMZ radio button in the **Data Source** dialog box and click **Browse**.

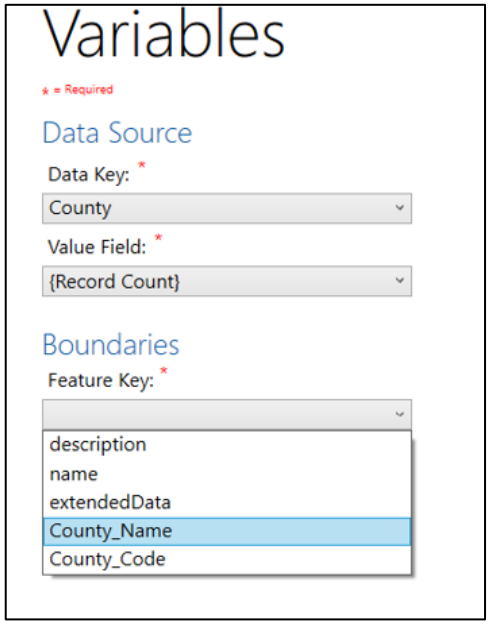


KML/KMZ button Data Source dialog

6. select the **Maryland_Counties.kml** file located in the **KML_Example** folder. Click **Open**.

7. In the **Data Key** drop-down, select **County.**

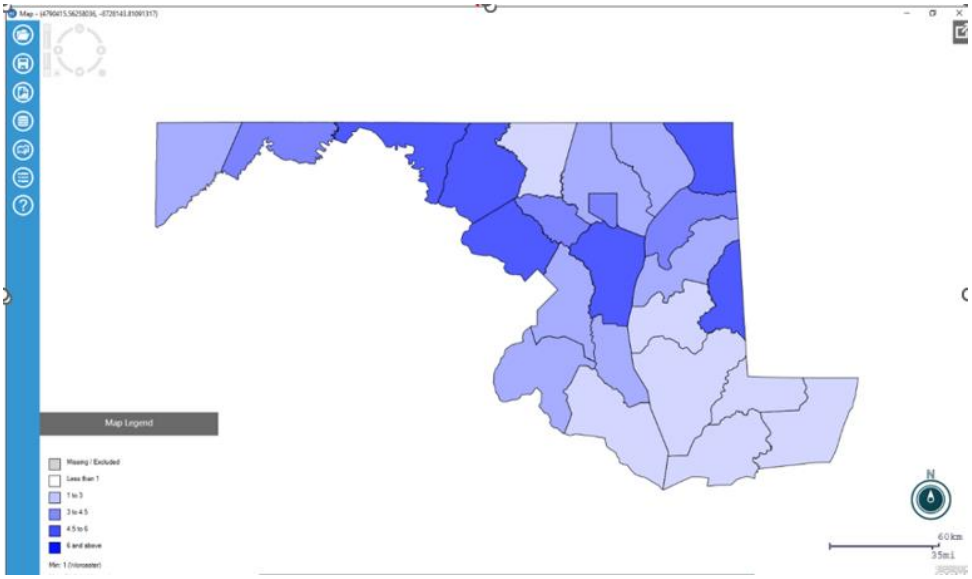
8. In the **Value Field** drop-down, select {Record Count. In the **Feature Key** drop-down, select County Name.



Data Source drop-down menu

9. In the **Display** property panel, set the desired display options such as legend title and text (labels). For this example, leave these options blank. You can also select the colors and data ranges to use, including the number of classes into which you would like to divide the data. For this example, leave the default selections. Click **OK.**

The choropleth map appears, showing the number of records in the dataset within each Maryland County. Counties with the highest tire of records are dark blue, while counties with the lowest number of records are light blue. Note that the "starting" data category is white, but no data fall into that category here.



Choropleth map from Lyme disease dataset

10. To change any of the display options or to add optional data filters, click the configuration icon in the map layer located on the bottom of the screen.

Dot Density Map using Shapefile Boundaries

To display densities across geographic boundaries, select **Dot Density** from the data layers list. Epi Map will populate the dot density map according to the dot value selected in step eight. Increasing the dot value increases the value each dot represents. The dot density map populates each dot randomly within a set of

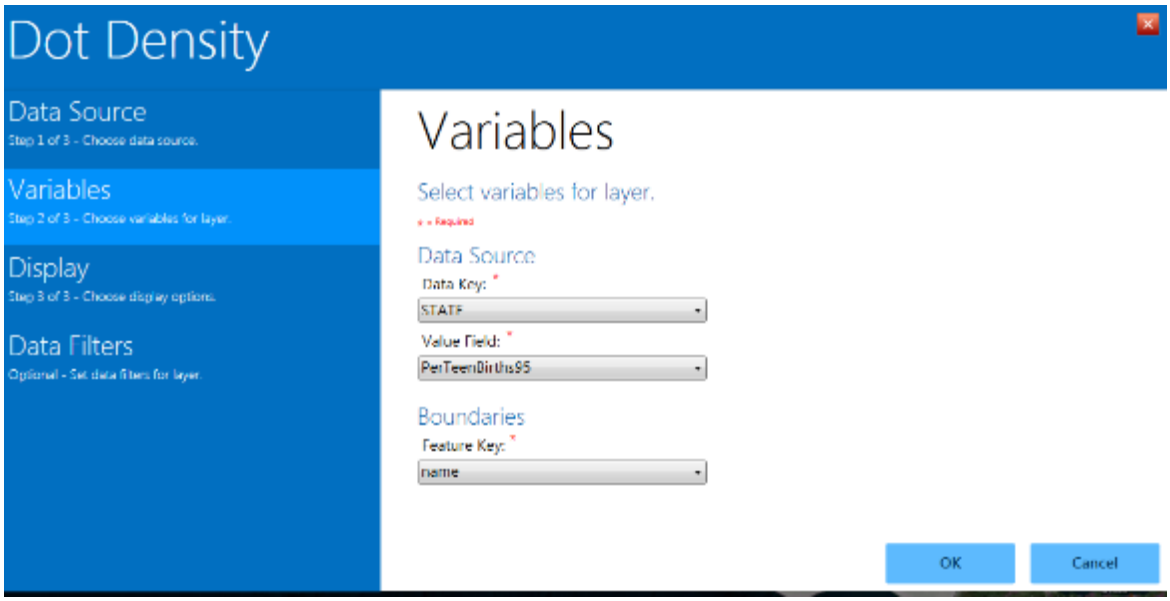
boundaries to display the density. The example below demonstrates how to create a dot density map with data from a Vital Statistics report published by the Mexico Ministry of Health using a shapefile.

1. Select **Add Data Layer** from the toolbar.
2. Select **Dot Density** from the list.

3. The **Data Source** dialog box opens.
4. Click **Browse** next to the Data Source field in the main layer dialog box, and then again in the **Data Source** dialog box. Select the **Sample** project folder and the Sample.prj file.
5. Select the **MexMap95** form in Data Source Explorer. Click **OK**.
6. Click **Browse** next to the **Shapefile** line. (If the **Browse** button is grayed out instead of blue, click the **Shapefile** radio button.)

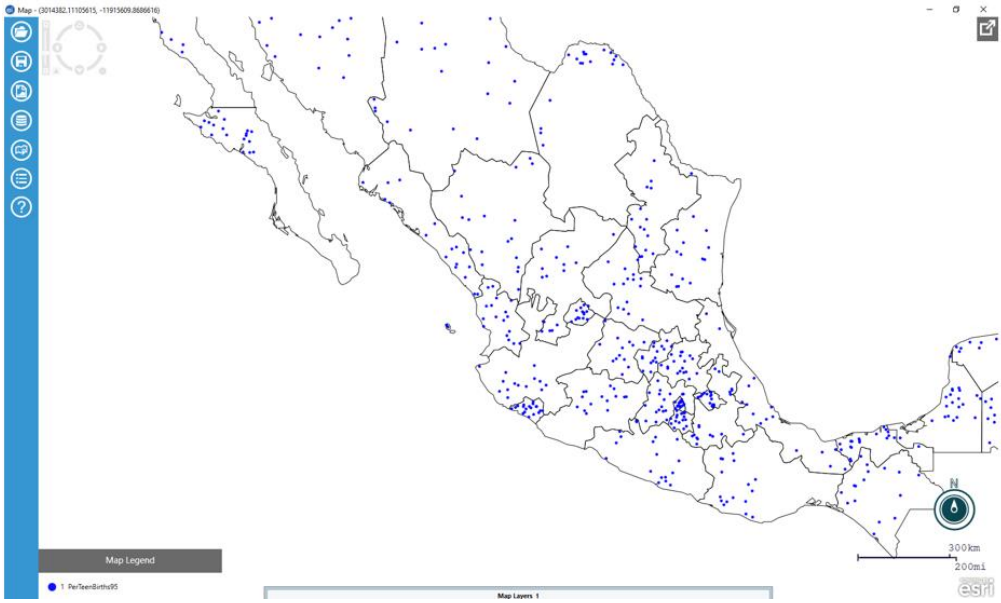
7. Select **MXState.shp** in the Sample.prj file.
8. Select **STATE** from the **Data Key** drop-down list. The data key connects the dataset field, STATE, to the shapefile feature key name.
9. Select **PerTeenBirths95** from the **Value Field** drop-down list. This populates the dot density map based on the percentage of teen births in that state.

Select **name** from the **Feature Key** drop-down list. The feature key connects name in the shapefile to the dataset.



Shapefile variable options and boundaries

10. Click **OK.**
11. The **Display** options dialog appears. Options can be selected here for the color of dots in the map and the value each dot should represent (**Dot Value**). In this example, the dots are associated with the percentage of teen pregnancy in each state. By specifying "1", each dot will represent a single teen birth percentage point.
12. Click **OK**.



Dot Density using shapefile

13. The **Dot Density** map data layer is displayed on the main Epi Map window. Since one (1) was selected as the dot value, each blue dot represents a single percentage point for the dataset. Areas with a higher concentration of red dots represent a higher percentage of teen births.
14. To change any of the display options or to add optional data filters, place the mouse over the Map Layers tab located at the bottom of the screen. When the tab scrolls up, click the layer configuration icon to re-open the **Choropleth** layer.



Figure Error! No text of specified style in document..**1: Dot Density Map Layers view and layer configuration icon**

Base Layer

About Base Layers

Base layers add extra information to your map beyond your project data. While you've already learned how to combine files to make color-shaded and dot-density maps, you might want to add more details like special locations or boundaries. You can add these extra layers using the **Add Base Layer** tool.

Error! No text of specified style in document.**2Add KML Files**

- KML files can show special features on your map. For example, you can see Hurricane Katrina's path:
- Click Add Base Layer in the toolbar
- Click the KML/KMZ File button
- Type or paste a URL for your KML file - like NOAA's Katrina tracker

  Your map will show the new information right away

Error! No text of specified style in document.**3Add Map Server Information**

Map servers give you access to official map information:

1. Click Add Base Layer in the toolbar

Error! No text of specified style in document..4 Click the **Map Server** button

3. Type or paste your Map Server URL

4. Pick what you want to see from the Features menu

When you **Error! No text of specified style in document.**5zoom in, you'll see more details appear on your **Error! No text of specified style in document.**6map.

Add Shapefiles

Shapefiles help you show boundaries, like state lines:

1. Click **Add Base Layer** in the toolbar

2. Click the **Shapefile** button

3. Click **Browse** to find your shapefile

1. 4. Pick the file you want (like **MXState.shp** for Mexican states)

4. Click **OK** to see your new boundaries on the map

Each kind of base layer adds different information to help make your map more useful. You can add these layers at any time **Error! No text of specified style in document.**7to show exactly what you need.



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